

3. Medical Image Segmentation

- 3.1 Taxonomy of segmentation algorithms
- 3.2 Validation of segmentation algorithms
- 3.3 Deterministic Filtering & Thresholding Approaches
- 3.4 Probabilistic Imaging Model
- 3.5 **Expectation Maximisation for GMM**
- 3.6 Image classification with bias field
- 3.7 Variational Bayes EM
- 3.8 STAPLE Algorithm

Expectation Maximisation Algorithm

- Iterative approach for estimating parameters of (Gaussian) Mixture parameters
- General Idea :
 - New criterion : Add unknown variable u (posterior) and add constraint (KL divergence)
 - Alternate maximization performed in closed form : equivalent to lower bound maximization

Alternate maximisation

- Replace Log-Likelihood with a criterion easier to optimize but with additional unknowns
- Log-(marginal) likelihood :

$$L(\theta) = \log \Lambda(\theta) = \sum_n \log p(x_n | \theta) = \sum_n \log(\sum_k \pi_k \mathcal{N}(x_n; \mu_k, \sigma_k))$$

- New criterion $F(\theta, u)$:
 - Add $u = \{u_{nk}\}$ as unknown. u is a vector of u_{nk} which is the posterior probability

$$F(\theta, u) = L(\theta) - D_{KL}(u || p(z|x))$$

- By maximizing F with respect to u ,

$$u_{nk} = p(z_{nk} = 1 | x_n)$$

Why is it easier to optimize $F(\theta, u)$?

- General result :
 - X = observed random variable
 - Z = hidden random variable
 - Joint probability $p(x_n, z_n) = p(x_n|z_n)p(z_n) = p(z_n|x_n)p(x_n)$
 - Constraint on u_{nk} : $\sum_k u_{nk} = 1$
 - Log likelihood : $L(\theta) = \sum_n \log p(x_n) = \sum_n \sum_k u_{nk} \log p(x_n)$
 - New criterion :

$$F(\theta, u) = \sum_n \sum_k u_{nk} \log p(x_n) - \sum_n \sum_k u_{nk} \log u_{nk}/p(z_{nk}|x_n)$$

$$F(\theta, u) = \sum_n \sum_k u_{nk} \log p(x_n, z_{nk}) - \sum_n \sum_k u_{nk} \log u_{nk}$$

We have « relaxed » the optimization problem by introducing a new unknown variable

Interpretation

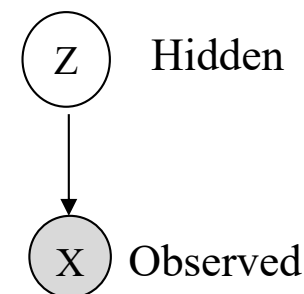
- New criterion involves 2 terms :

$$F(\theta, u) = \underbrace{\sum_n \sum_k u_{nk} \log p(x_{nk}, z_{nk})}_{Q(\theta, u)} - \underbrace{\sum_n \sum_k u_{nk} \log u_{nk}}_{\mathbb{H}(u)}$$

- $F(\theta, u)$ is the *variational lower bound*
- $-F(\theta, u)$ is the *variational free energy* = average energy - entropy
- $Q(\theta, u) = \sum_n \sum_k u_{nk} \log p(x_{nk}, z_{nk}) = \mathbb{E}_U(\log p(X, Z))$ is the expectation of the complete likelihood
- $\mathbb{H}(u) = -\sum_n \sum_k u_{nk} \log u_{nk}$ is the **entropy** of the approximate posterior probability
- $Q(\theta, u)$ is easier to optimize wrt θ because it involves complete likelihood = likelihood of observed and hidden variables

Evidence Lower Bound

- General result :
 - For any inverse problem where Z is the hidden variable and X observed variable :



$$\begin{aligned} \log p(X) - D_{KL}(u || p(Z|X)) \\ = \mathbb{E}_u(\log p(X, Z)) + \mathbb{H}(u) \end{aligned}$$

- Variational lower bound :
$$\log p(X) \geq \mathbb{E}_u(\log p(X, Z)) + \mathbb{H}(u)$$

Case of Gaussian Mixtures

- Log likelihood

$$L(\theta) = \log \Lambda(\theta) = \sum_n \log(\sum_k \pi_k \mathcal{N}(x_n; \mu_k, \sigma_k))$$

- Function of parameters :

$$Q(\theta, u) = \sum_n \sum_k u_{nk} \log \pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k)$$

- Note that we have sum of log instead of log of sums !
- Criterion $F(\theta, u) = Q(\theta, u) + \mathbb{H}(u)$ is known as **Hathaway criterion**

EM Algorithm

- The algorithm optimizes alternatively between u and θ = coordinate ascent

$$F(\theta, u) = L(\theta) - D_{KL}(u || p(z|x)) = Q(\theta, u) + \mathbb{H}(u)$$

- Constraints : $\sum_k \pi_k = 1 \quad \sum_k u_{nk} = 1$
- E-step

- maximize $F(\theta, u)$ wrt u

Compute $u_{nk} = \frac{\pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k)}{\sum_{k=1}^K \pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k)}$

- Equivalent to minimizing KL divergence between u and posterior probability

M-Step

- M-step : maximize $F(\theta, u)$ or equivalently $Q(\theta, u)$ wrt $\theta = \{\theta_S, \theta_I\}$

- Optimize with respect to mean μ_k

$$\frac{\partial Q}{\partial \mu_k} = 0$$



$$\mu_k = \frac{\sum_{n=1}^N u_{nk} x_n}{\sum_{n=1}^N u_{nk}}$$

- Optimize with respect to covariance Σ_k

$$\frac{\partial Q}{\partial \Sigma_k} = 0$$



$$\Sigma_k = \frac{\sum_{n=1}^N u_{nk} (x_n - \mu_k)(x_n - \mu_k)^T}{\sum_{n=1}^N u_{nk}}$$

- Optimize with respect to prior probabilities

$$\frac{\partial Q}{\partial \pi_k} = 0$$



$$\pi_k = \frac{1}{N} \sum_{n=1}^N u_{nk}$$

EM Algorithm for GMM

- Iterative scheme
 - Make initial guesses for the parameters
 - Alternate between the following two stages:
 1. E-step: evaluate posterior u_{nk}

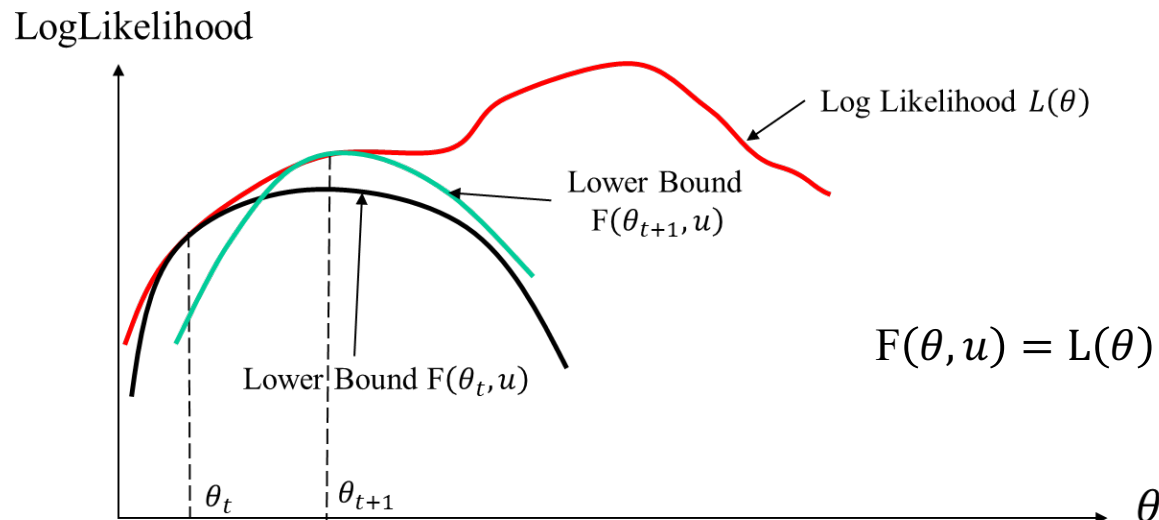
$$u_{nk} = \frac{\pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k)}{\sum_{k=1}^K \pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k)}$$

2. M-step: update parameters (μ_k, Σ_k, π_k) using ML results

$$\mu_k = \frac{\sum_{n=1}^N u_{nk} x_n}{\sum_{n=1}^N u_{nk}} \quad \Sigma_k = \frac{\sum_{n=1}^N u_{nk} (x_n - \mu_k)(x_n - \mu_k)^T}{\sum_{n=1}^N u_{nk}} \quad \pi_k = \frac{1}{N} \sum_{n=1}^N u_{nk}$$

EM as Iterated Lower Bound Maximisation

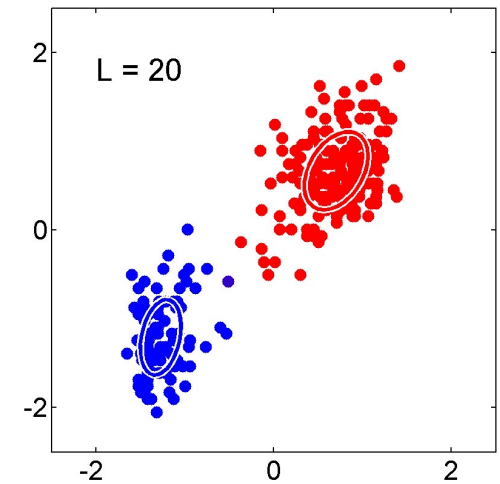
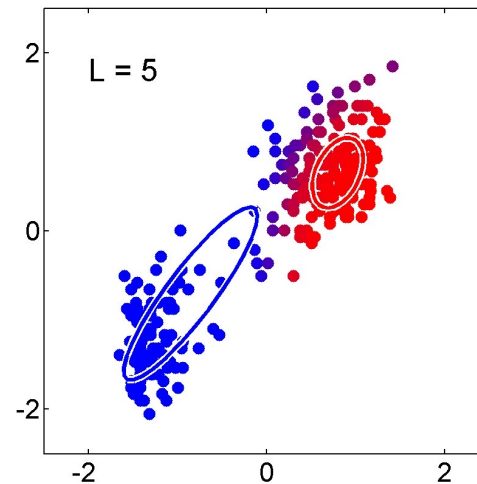
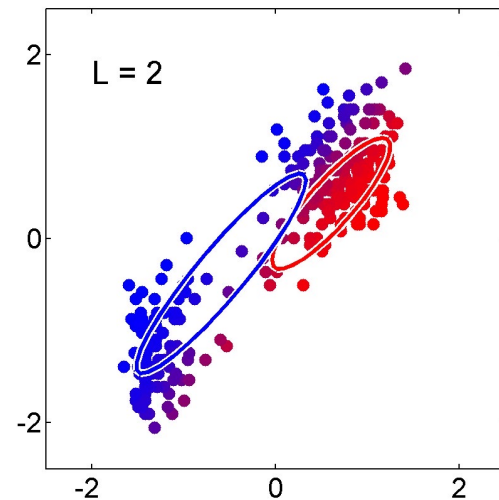
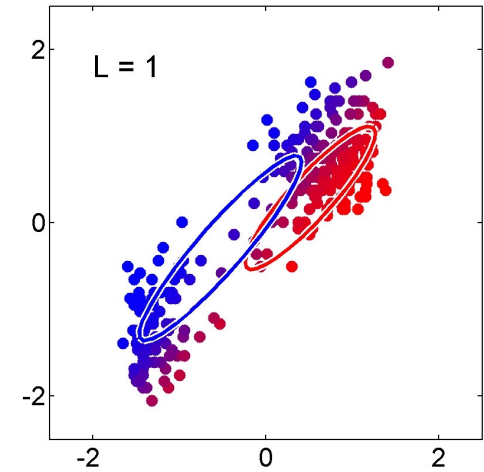
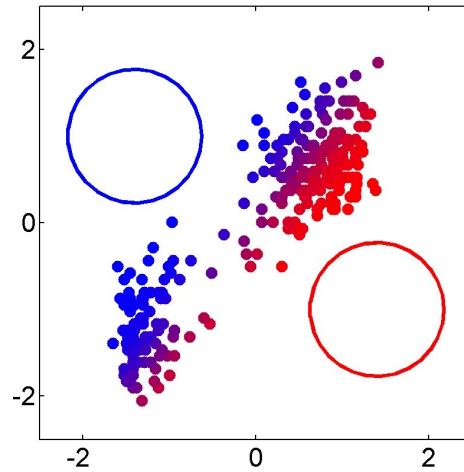
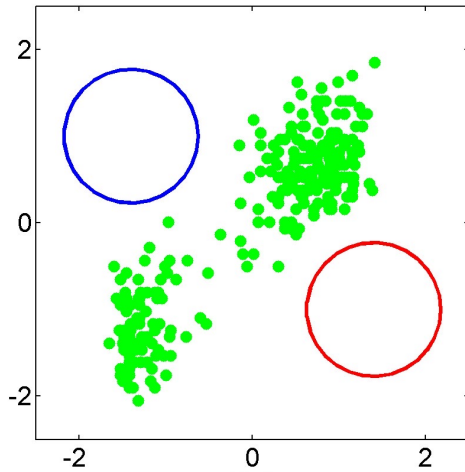
- Equivalent view of EM algorithm :
 - E-step leads to $u = p(z|x)$ and therefore makes $L(\theta_t) = F(\theta_t, u)$.
 - $F(\theta, u)$ is a lower bound of Log-likelihood $L(\theta)$ since Kullback Leibler divergence is positive
 - M-step optimizes $F(\theta, u)$ with respect to θ which is easier to maximize than log likelihood



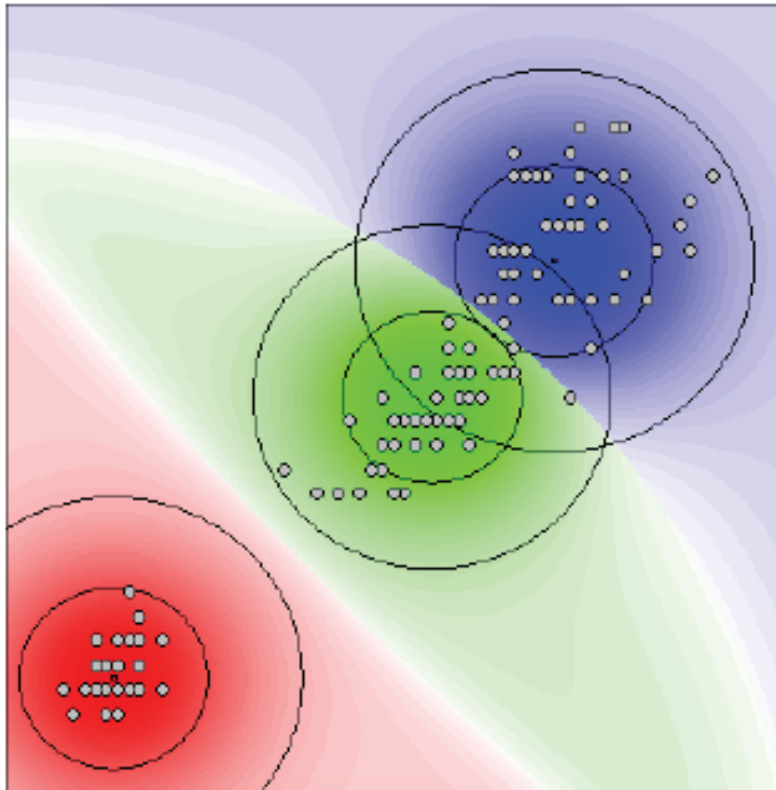
$$L(\theta) \geq F(\theta, u)$$

$$F(\theta, u) = L(\theta) - D_{KL}(u || p(z|x)) = Q(\theta, u) + \mathbb{H}(u)$$

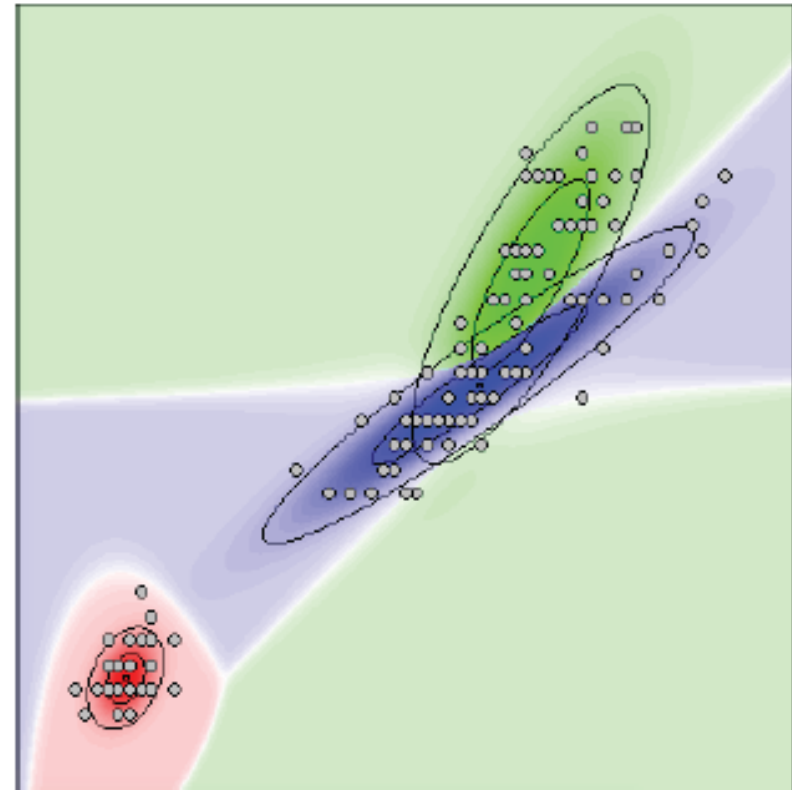
Example of EM with 2 Gaussian distributions



EM on Iris data



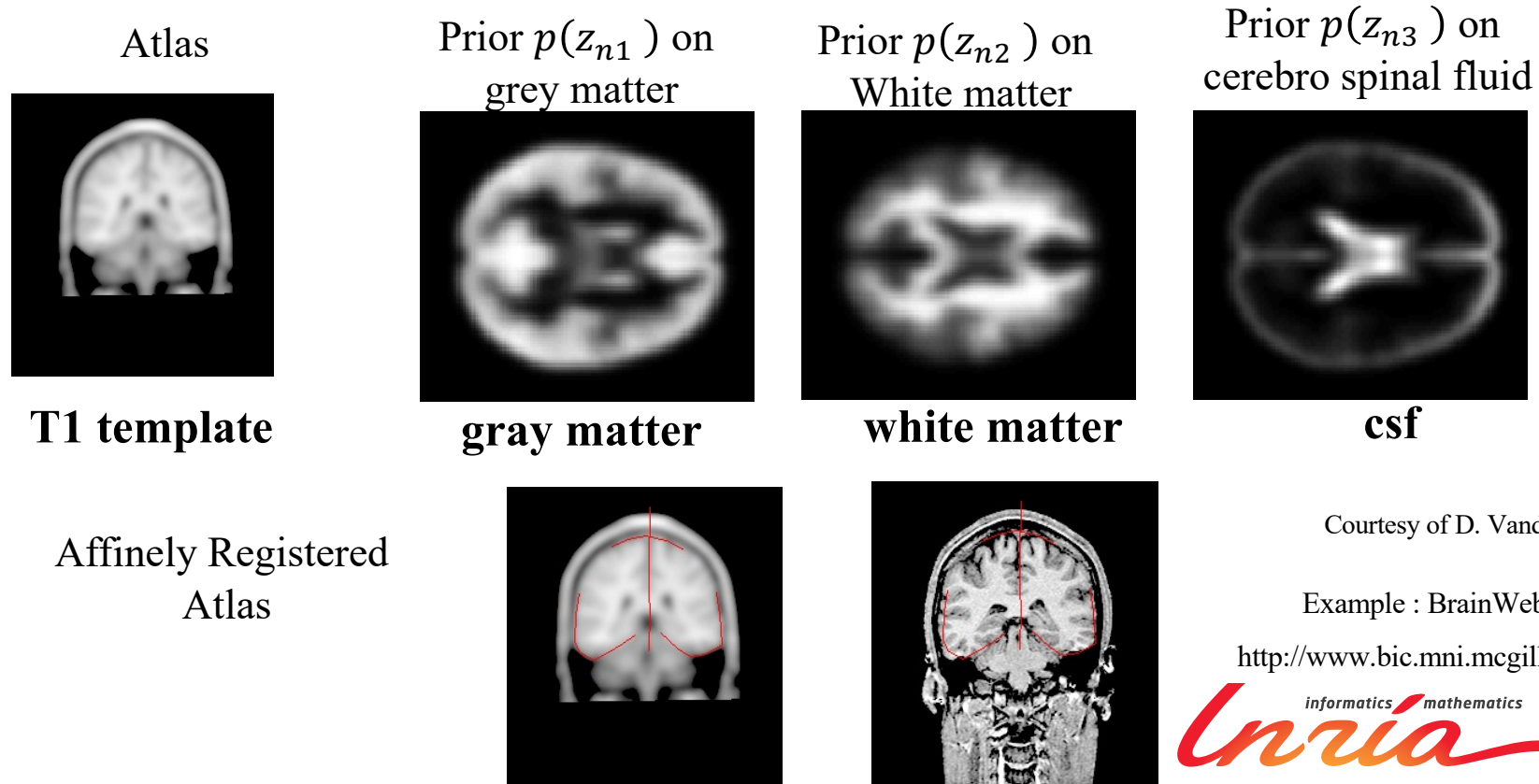
equal prior, spherical



equal prior, ellipsoidal

Class Priors

- Initial hypothesis : homogeneous priors $p(z_{nk} = 1) = \pi_k$ is estimated
- Priors may be given by atlas registered on images. In this case θ_S are the registration parameters



Courtesy of D. Vandermeulen

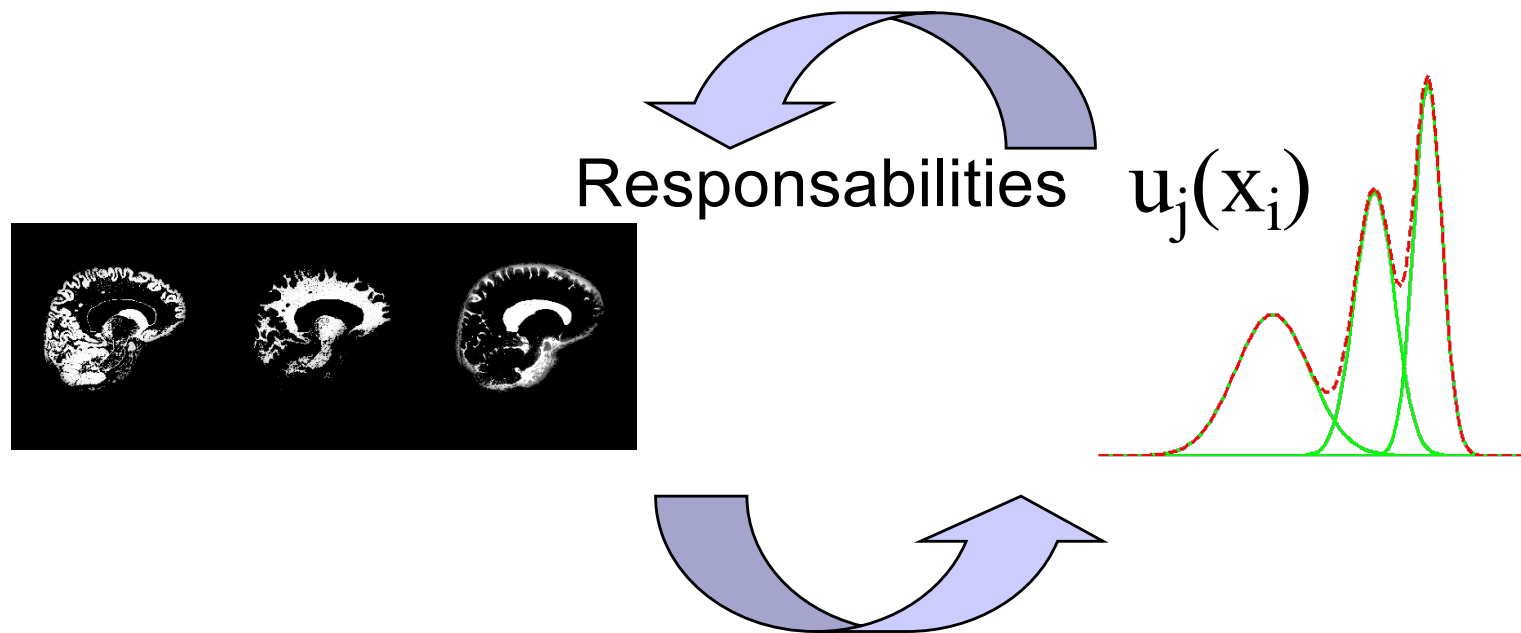
Example : BrainWeb at MNI

<http://www.bic.mni.mcgill.ca/brainweb/>

EM for Image Intensity Classification

- Use the EM algorithm
[Dempster77,Wells94] :

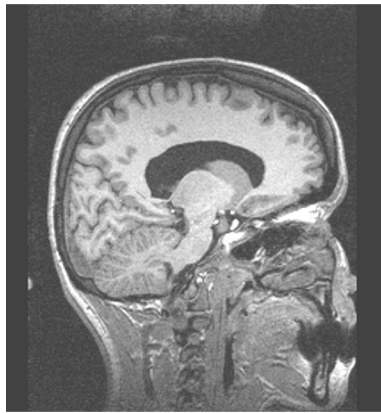
Expectation-Maximisation



Mixture Param. μ_K Σ_K π_K

Brain Tissue Classification

- Typical application : use MR cerebral



Courtesy of D. Vandermeulen



Cerebro-spinal fluid

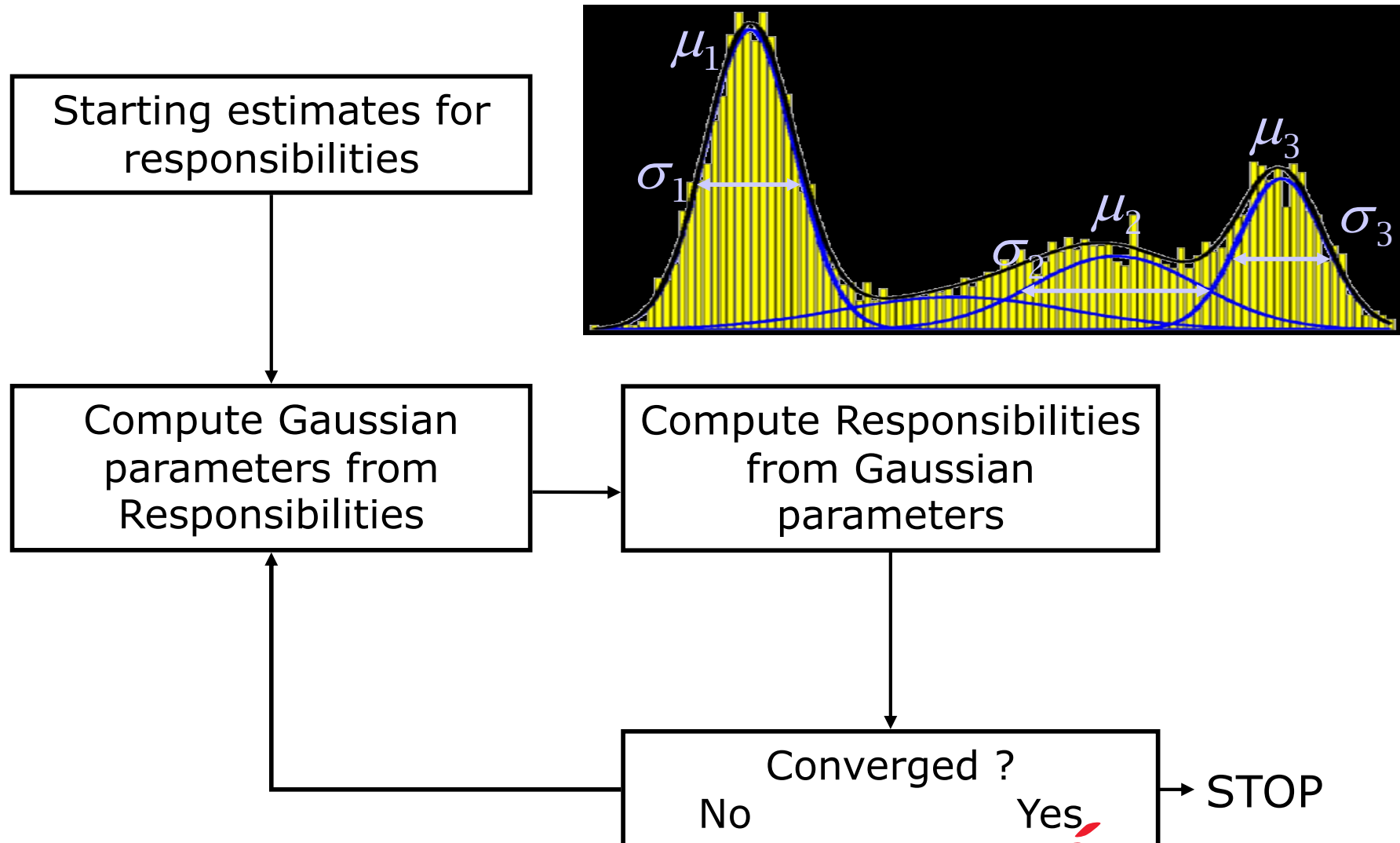
Grey matter

White matter

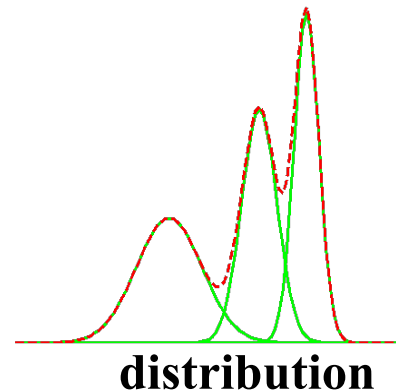
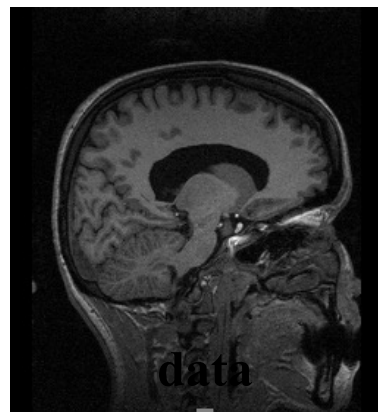
3 Classes

Scalar feature
= Intensity

EM Classification - Algorithm



Stage 1: Expectation



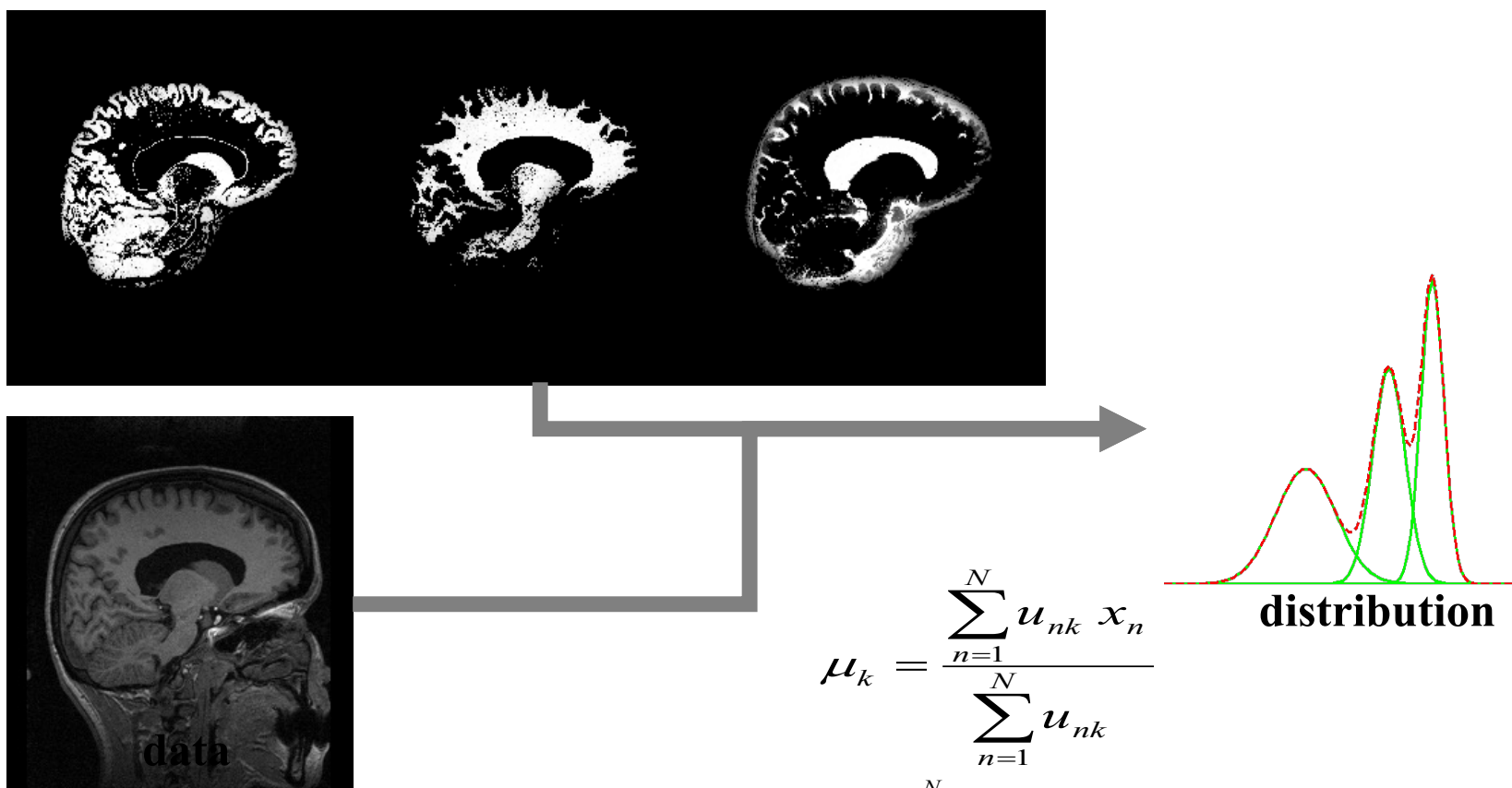
Compute
Responsibilities

$$u_{nk} = \frac{\pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k)}{\sum_{k=1}^K \pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k)}$$



Courtesy of D. Vandermeulen

Stage 2: Maximization



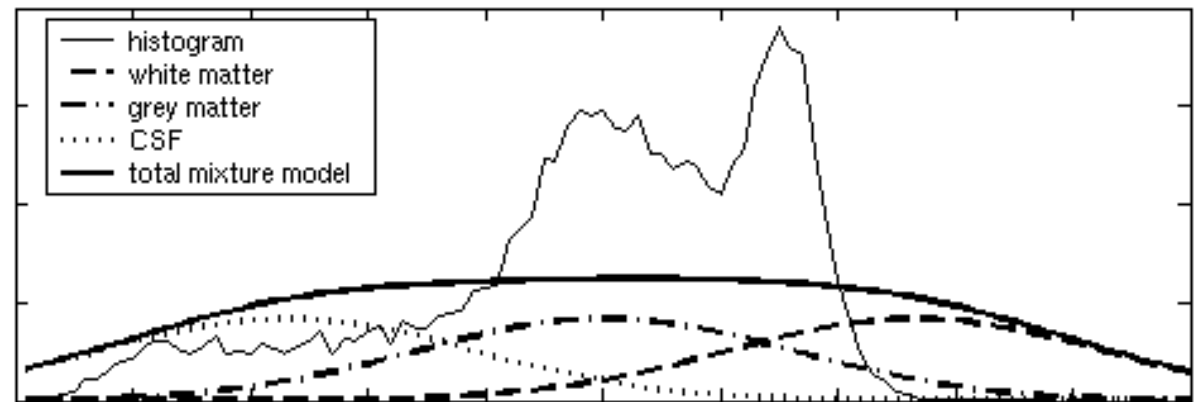
Courtesy of D. Vandermeulen

$$\mu_k = \frac{\sum_{n=1}^N u_{nk} x_n}{\sum_{n=1}^N u_{nk}}$$

$$\Sigma_k = \frac{\sum_{n=1}^N u_{nk} (x_n - \mu_k)(x_n - \mu_k)^T}{\sum_{i=1}^N u_{nk}}$$

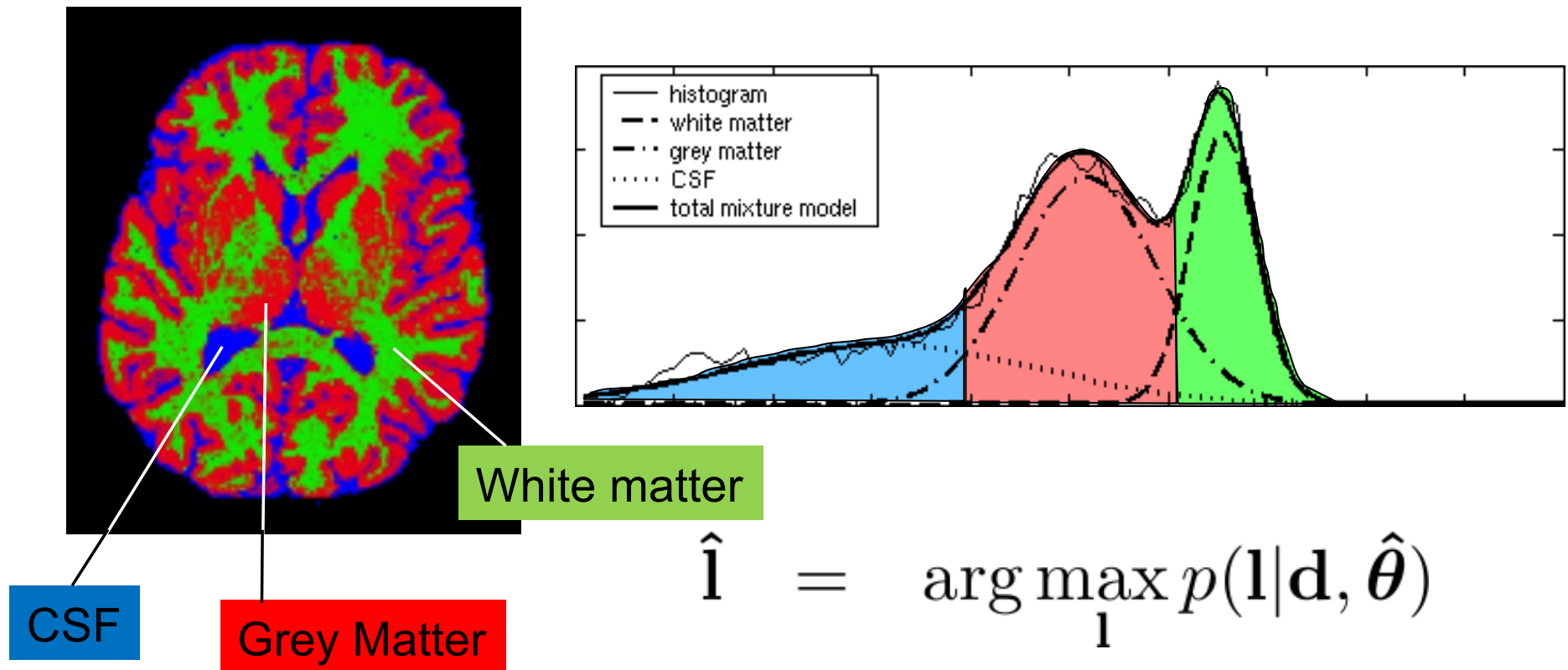
$$\pi_k = \frac{1}{N} \sum_{n=1}^N u_{nk}$$

Iterations EM



Courtesy of K. Van Leemput

Results



Courtesy of K. Van Leemput

GMM and K-Means

- GMM with :
 - Isotropic variance $\Sigma_k = \epsilon Id$
 - Uniform prior : $\pi_k = \frac{1}{K}$
- Expectation of complete Lik. : $Q(\theta) = - \sum_n \sum_k \frac{u_{nk} |x_n - \mu_k|^2}{2\epsilon}$
- Same as Fuzzy-Cmeans with m=1
- Same as K-means when :
 - $\epsilon \rightarrow 0$
 - $u_{nk} \in \{0,1\}$

$$u_{nk} = \frac{\exp(-|x_n - \mu_k|^2 / 2\epsilon)}{\sum_{j=1}^K \exp(-|x_n - \mu_j|^2 / 2\epsilon)} \rightarrow r_{nk} \in \{0,1\}$$

K Means functional

- K Means algorithm consists in optimizing the functional :
 - $J(r, \mu) = \sum_{n=1}^N \sum_{k=1}^K r_{nk} \|x_n - \mu_k\|^2$
 - With the constraint that $r_{nk} \in \{0,1\}$ and $\sum_{k=1}^K r_{nk} = 1 \quad \forall n$
- J can be seen as
 - minimizing the correlation between the assignment and the distance to cluster center
 - Minimizing the compactness of the clusters

K Means optimization

- Perform alternate optimization :
 - Consider μ_k fixed and optimize on r_{nk}
 - For each data x_n choose which r_{nk} is 1

E-Step

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg \min_j \|x_n - \mu_k\| \\ 0 & \text{otherwise} \end{cases}$$

- Consider r_{nk} fixed and optimize on μ_k

M-Step

$$\frac{\partial J}{\partial \mu_k} = 2 \sum_{n=1}^N r_{nk} (\mu_k - x_n) = 0$$

$$\mu_k = \frac{\sum_{n=1}^N r_{nk} x_n}{\sum_{n=1}^N r_{nk}}$$

Good Initial Seeds (kmeans++)

- Choose the centers as far away as possible from each other but in a random manner.
- Algorithm :
 - Choose one center at random μ_1
 - While $k \leq K$
 - Compute $d_n = \arg \min_{j < k} \|x_n - \mu_j\|^2$ the minimum distance of data x_n to the already chosen centers
 - Pick μ_k among data with probability proportional to d_n
 - $k=k++$

David Arthur and Sergei Vassilvitskii. k-means++: The advantages of careful seeding. "Proceedings of the eighteenth annual ACM-SIAM symposium on Discrete algorithms", 2007 , pp. 1027–1035

Issues with EM for GMM

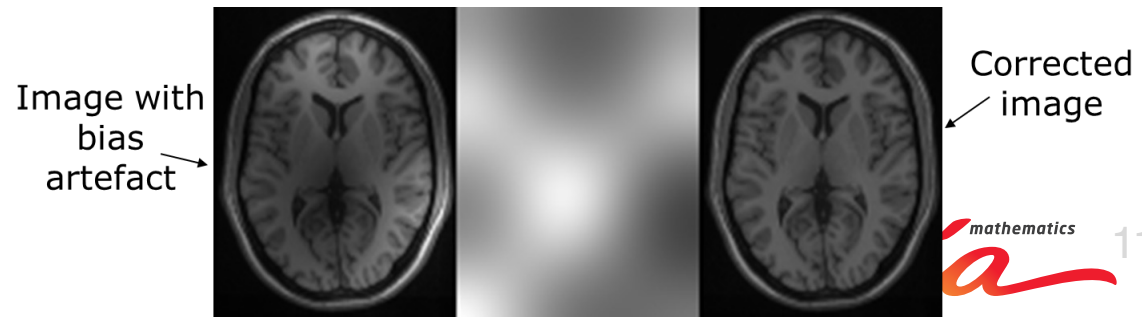
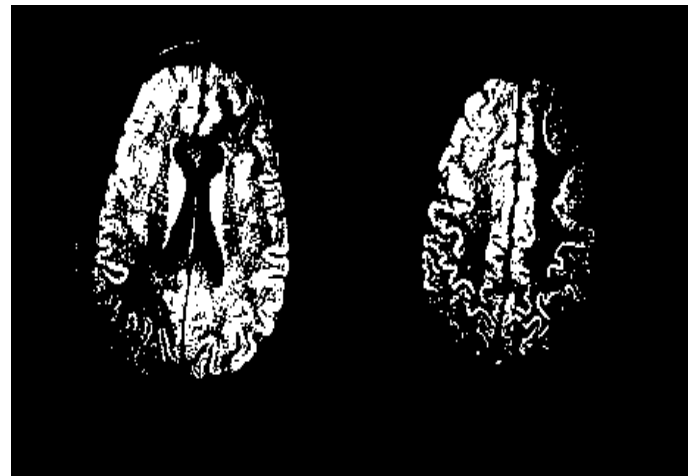
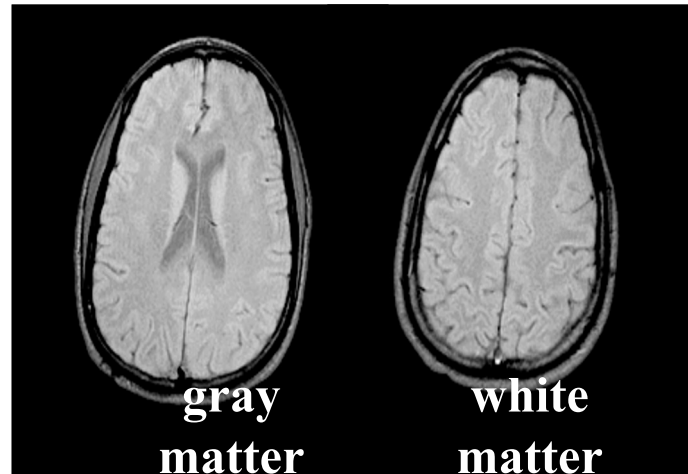
- Presence of bias field in MR images
- EM leads to only local maxima of Log-likelihood
- Functional admits trivial solutions (zero covariance centered at data points) that can lead to bad estimate
- The covariance matrix Σ_k should be invertible which is not guaranteed (may use pseudo-inverse)
- How to choose the number of classes
- How to make the estimation robust to outliers ?

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Issue : presence of MR bias field

MR images are corrupted by a smooth intensity non-uniformity (bias).



MR bias field estimation methods

- homogeneous (water) phantom measurements
 - non-retrospective
 - bias is patient-dependent
 - same MRI parameter settings for patient and phantom
- analytic correction of antenna receptor profile (idem)
- bias field estimation helps segmentation => segmentation helps bias field estimation?
 - Dawant et al: manual selection (or 1 iter.) of WM points + LSQ spline fit
 - Meyer et al.: region based, too many degrees of freedom
 - Wells et al.: EM-based estimation of bias+classification, requires pre-set MRI intensity model-parameters

Biased Gaussian Mixture model

- Bias field is modeled as a additive or multiplicative Noise
- Bias field parameters with smooth linear combination of smooth basis functions

C_k are the parameters controlling the bias field

$$b(x) = \sum_{l=1}^M C_l \phi_l(r(x))$$

$r(x)$ is the position of voxel of intensity x

- Convenient choice : additive noise (but not realistic)

$$p(x) = \sum_k \pi_k \mathcal{N}\left(x - \sum_{l=1}^M C_l \phi_l(r(x)) \mid \mu_k, \sigma_k\right)$$

- Use Log of image to cope with additive noise

Bias Description

- Bias is described as a combination of slowly varying polynomials :

$$b(x) = \sum_{l=1}^M C_l \phi_l(r(x))$$

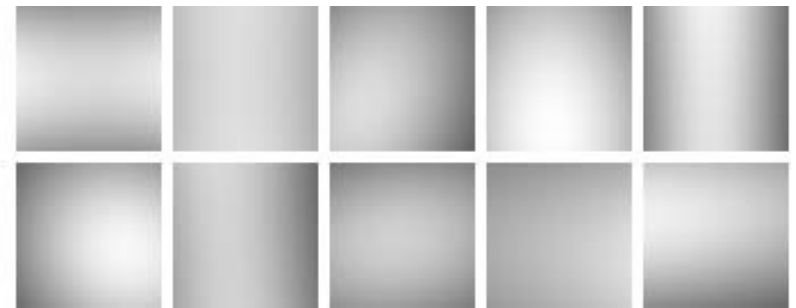
$r(x)$ is the position of voxel x

C_l are the parameters controlling the bias field

- $\phi_l(r(x))$ is a slowly varying polynomial :

- For instance :

- $\phi_0(x,y,z)=1$
- $\phi_1(x,y,z)=x-t_x/2$
- $\phi_2(x,y,z)=(y-t_y/2)$
- $\phi_3(x,y,z)=(z-t_z/2)$
- $\phi_4(x,y,z)=(x-t_x/2)*(y-t_y/2)$
-



Bias field with brain mask

Bias Description

- Bias field is modeled as an additive or multiplicative noise

$$b(x) = \sum_{l=1}^M c_l \phi_l(r(x))$$

- Multiplicative Noise :



No closed form solution

$$p(x|\theta) = \sum_k \pi_k \mathcal{N} \left(x * \left(\sum_{l=1}^M c_l \phi_l(r(x)) \right) | \mu_k, \sigma_k \right)$$

- Additive Noise using the Log of the image intensity:



closed form solution

$$p(x|\theta) = \sum_k \pi_k \mathcal{N} \left(x - \sum_{l=1}^M c_l \phi_l(r(x)) | \mu_k, \sigma_k \right)$$

Extended EM Algorithm

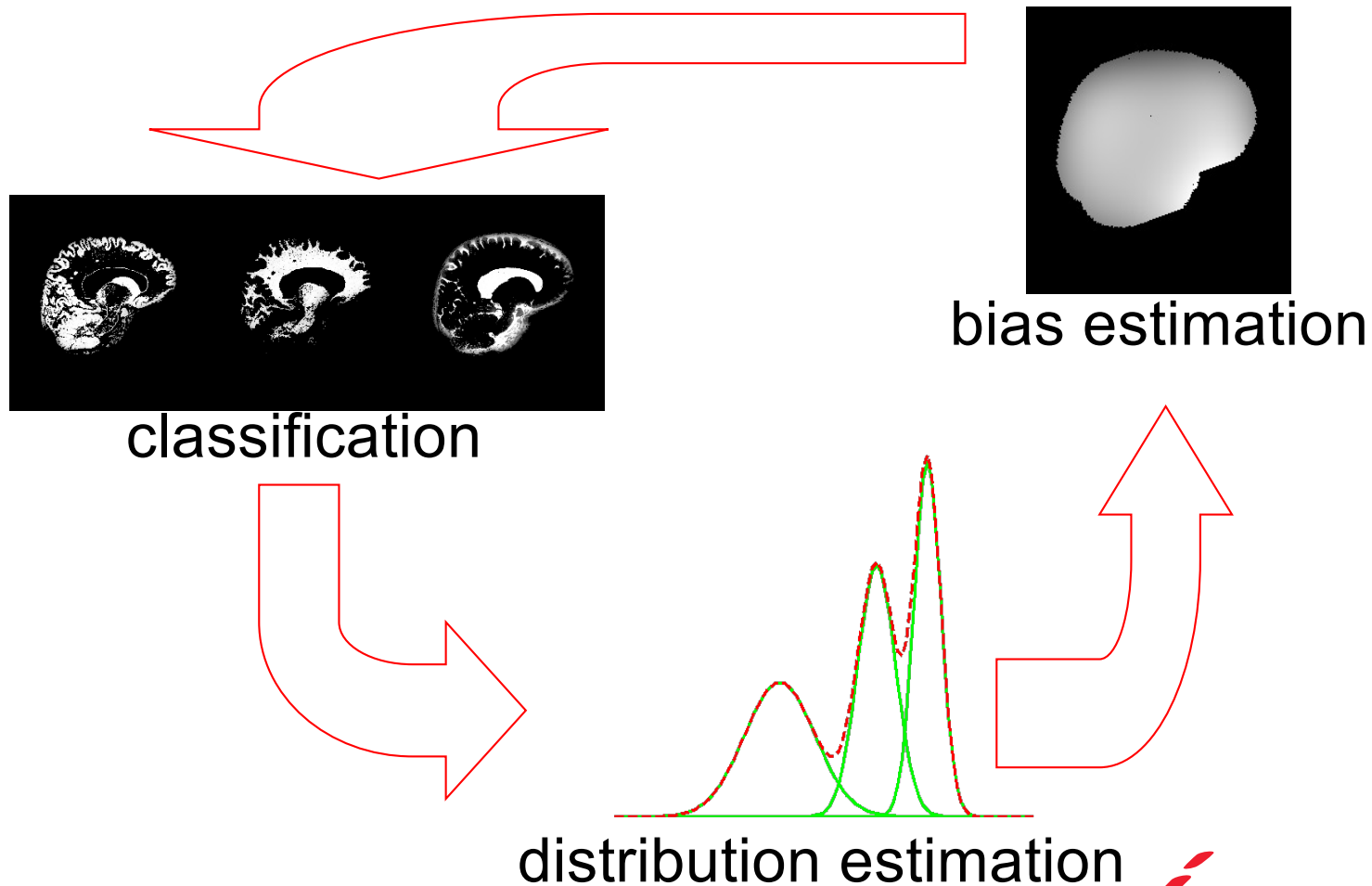
- Include new parameters
 - Bias description $C=\{C_l\}$
- Define Extended $Q(u, \theta) = \mathbb{E}_U(\log p(X, Z))$

$$Q(u, \theta) = \sum_n \sum_k \log \left(u_{nk} \mathcal{N} \left(x_n - \sum_{l=1}^M C_l \phi_l(r_n) \mid \mu_k, \Sigma_k \right) \right) \quad r_n = \text{position of voxel } n$$

- Define new posterior probabilities :

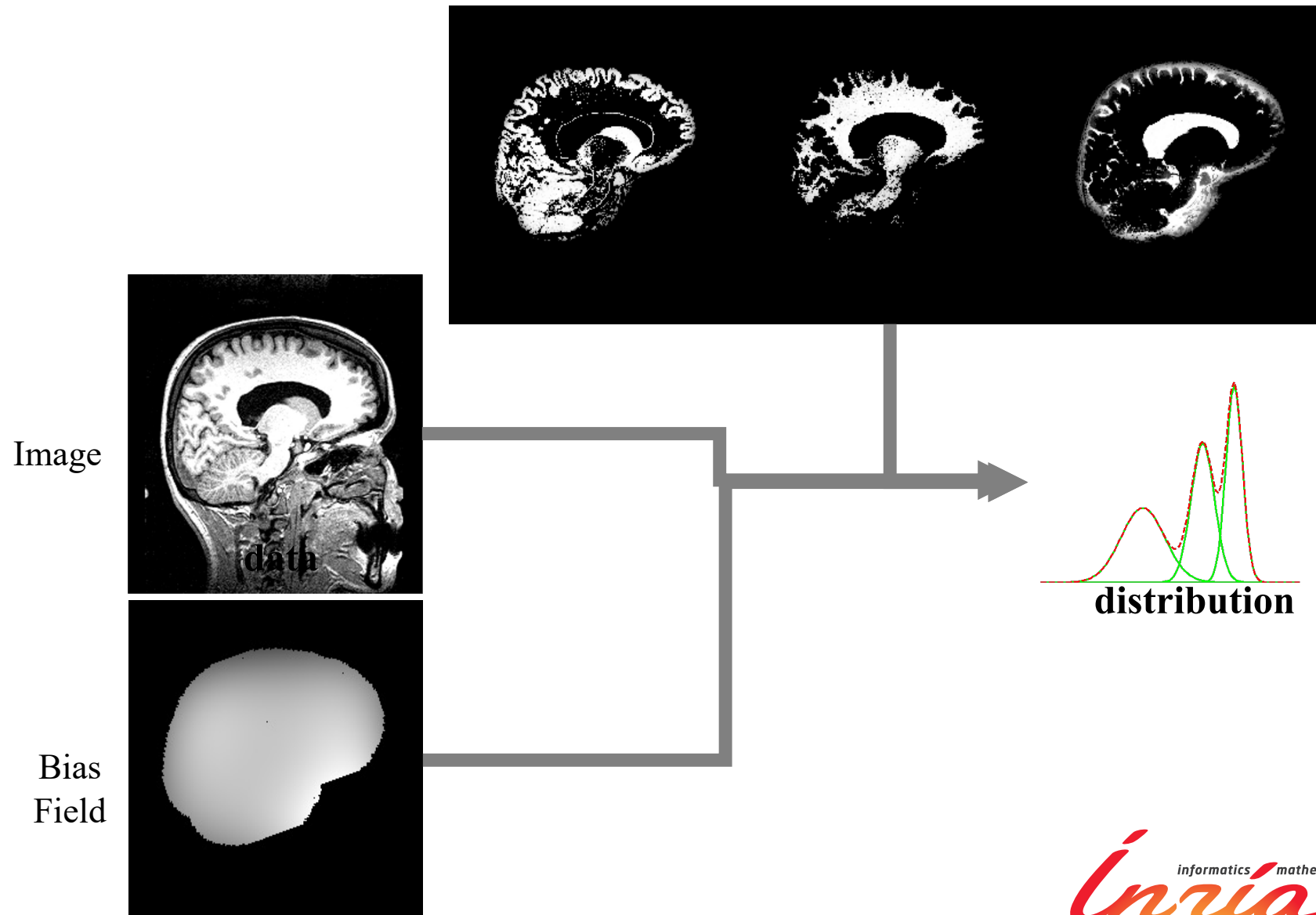
$$u_{nk} = p(z_{nk} = 1 \mid \theta, C) = \frac{\pi_k \mathcal{N} \left(x_n - \sum_{l=1}^M C_l \phi_l(r_n) \mid \mu_k, \Sigma_k \right)}{\sum_j^K \pi_j \mathcal{N} \left(x_n - \sum_{l=1}^M C_l \phi_l(r_n) \mid \mu_j, \Sigma_j \right)}$$

Extended EM algorithm



Step 1: distribution estimation

Belonging Probabilities



Step1 : distribution estimation

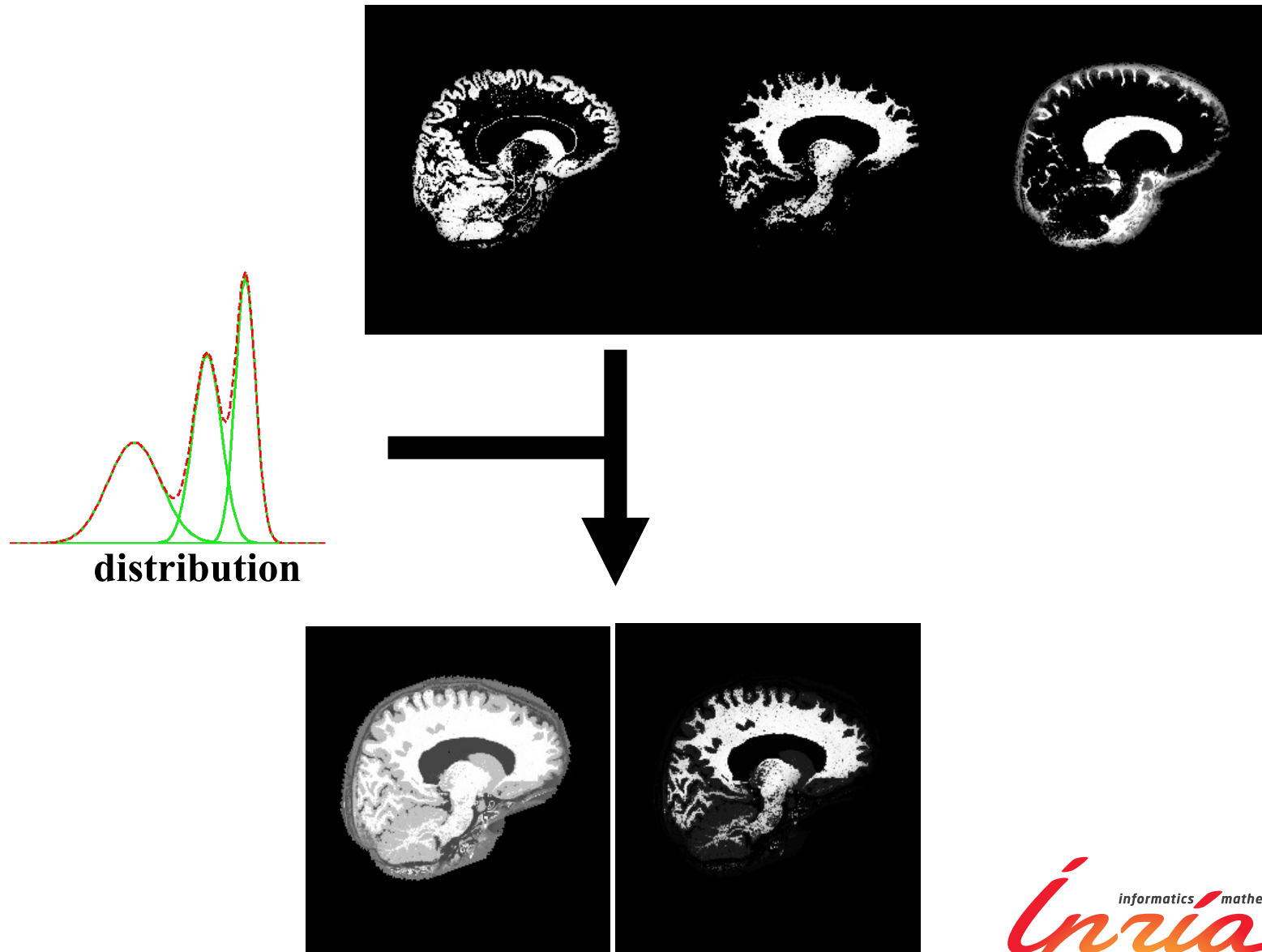
- Compute mean and variance based on the bias corrected intensity

$$\mu_k = \frac{\sum_{n=1}^N u_{nk} (x_n - \sum_{l=1}^M C_l \phi_l(r_n))}{\sum_{n=1}^N u_{nk}}$$

$$\Sigma_k = \frac{\sum_{n=1}^N u_{nk} (x_n - \sum_{l=1}^M C_l \phi_l(r_n) - \mu_k)^2}{\sum_{n=1}^N u_{nk}}$$

Step 2: bias field estimation

Belonging Probabilities



Reminder :

Linear Least Square Problems

- Linear Least Square :
 - For any rectangular matrix X , Find X such that $\hat{X} = \arg \min_X \|AX - B\|^2$
 - If $\text{rank}(A)$ is full then $\hat{X} = (A^T A)^{-1} A^T B$
 $A^+ = (A^T A)^{-1} A^T$ is the pseudo-inverse
- Weighted Least Square
 - For any diagonal matrix W Find X such that $\hat{X} = \arg \min_X (AX - B)^T W (AX - B)$
 - If $\text{rank}(A)$ is full then $\hat{X} = (A^T W A)^{-1} A^T W B$

Bias Field Estimation

- Estimate Bias Field to minimize :

$$Q(\theta, C) = -\sum_n \sum_k u_{nk} \log(\pi_k \mathcal{N}(x_n - \sum_l C_l \phi_l(r_n); \mu_k, \Sigma_k)) + \dots$$

$$Q(\theta, C) = -\sum_n \sum_k u_{nk} \left(\frac{x_n - \sum_l C_l \phi_l(r_n) - \mu_k}{2\Sigma_k} \right)^2 + \dots$$

- C solution of a weighted least square problem

$$Q(\theta, C) = (AC - R)^T W (AC - R) + \dots$$

Bias Field Estimation

- Estimate Bias Field in the least square sense

$$\begin{bmatrix} C_1 \\ \vdots \\ C_M \end{bmatrix} = (A^T W A)^{-1} A^T W R$$

where

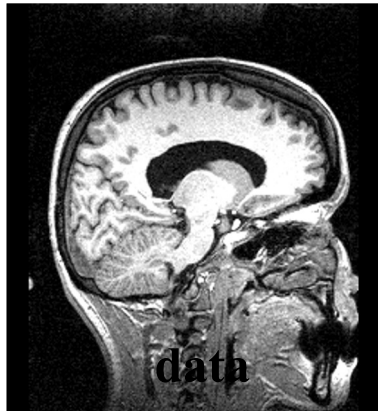
$$A = \begin{bmatrix} 1 & \dots & \phi_M(r_1) \\ 1 & \dots & \vdots \\ 1 & \dots & \phi_M(r_n) \end{bmatrix}$$

$$W = \begin{bmatrix} W_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & W_N \end{bmatrix}$$

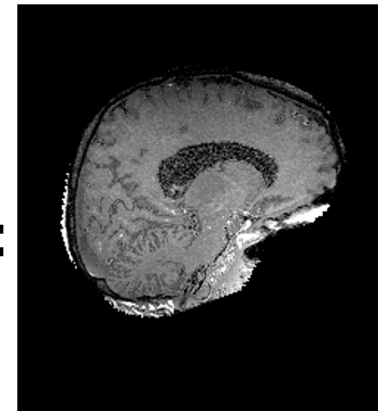
$$W_n = \sum_{j=1}^K \frac{u_{nj}}{\Sigma_j}$$

Difference between
Intensity and expected intensity
without any bias

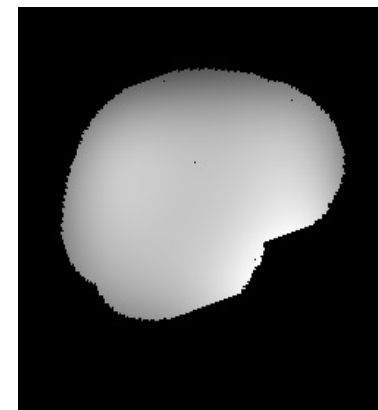
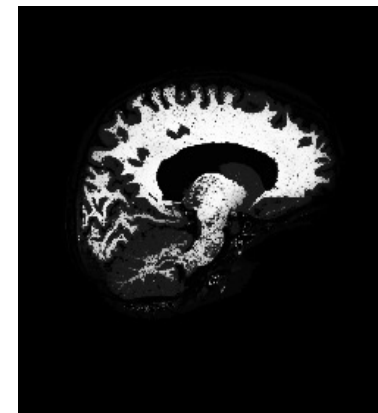
$$R = \begin{bmatrix} x_1 - \frac{\sum_{k=1}^K u_{1k} \mu_k / \Sigma_k}{\sum_{k=1}^K u_{1k} / \Sigma_k} \\ x_2 - \frac{\sum_{k=1}^K u_{2k} \mu_k / \Sigma_k}{\sum_{k=1}^K u_{2k} / \Sigma_k} \\ \vdots \\ x_M - \frac{\sum_{k=1}^K u_{Mk} \mu_k / \Sigma_k}{\sum_{k=1}^K u_{Mk} / \Sigma_k} \end{bmatrix}$$



=

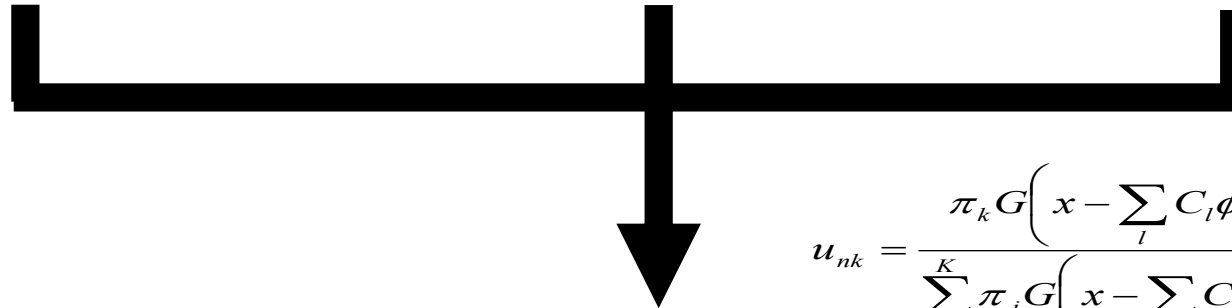
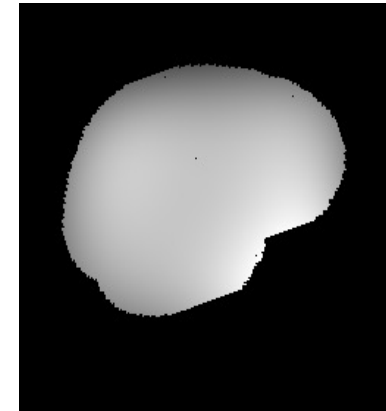
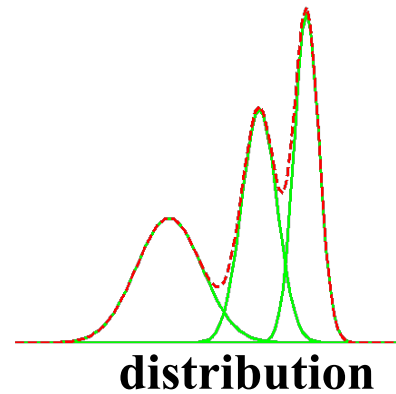
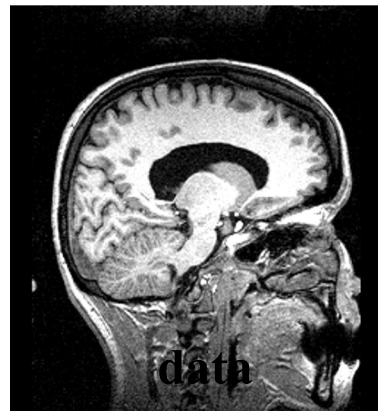


$$\begin{bmatrix} C_1 \\ \vdots \\ C_M \end{bmatrix} = (A^T W A)^{-1} A^T W R$$



$$R = \begin{bmatrix} x_1 - \frac{\sum_{k=1}^K u_{1k} \mu_k / \Sigma_k}{\sum_{k=1}^K u_{1k} / \Sigma_k} \\ x_2 - \frac{\sum_{k=1}^K u_{2k} \mu_k / \Sigma_k}{\sum_{k=1}^K u_{2k} / \Sigma_k} \\ \vdots \end{bmatrix}$$

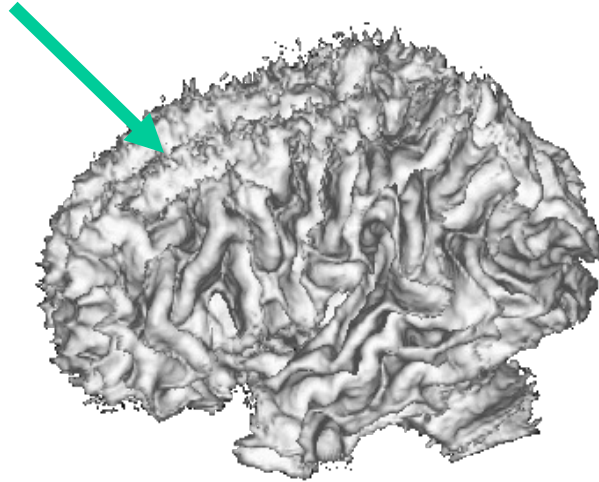
Step 3: classification



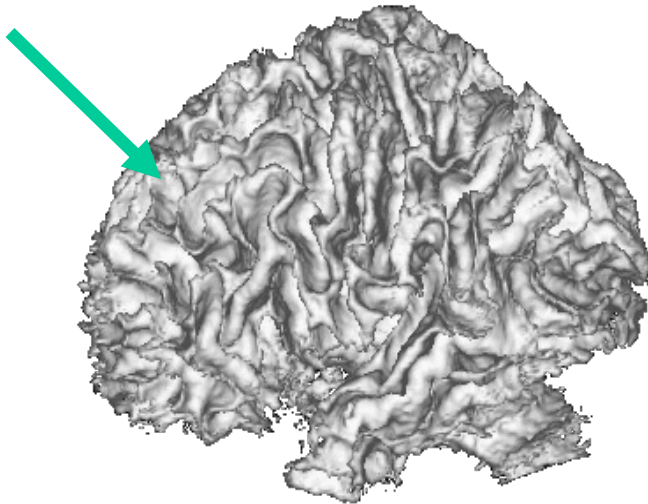
$$u_{nk} = \frac{\pi_k G\left(x - \sum_l C_l \phi_l(r_n) \mid \mu_k, \Sigma_k\right)}{\sum_{j=1}^K \pi_j G\left(x - \sum_l C_l \phi_l(r_n) \mid \mu_j, \Sigma_j\right)}$$



Results



**without
bias correction**



**with
bias correction**



white matter surface

Courtesy of D. Vandermeulen

gray matter surface
Inria

3. Medical Image Segmentation

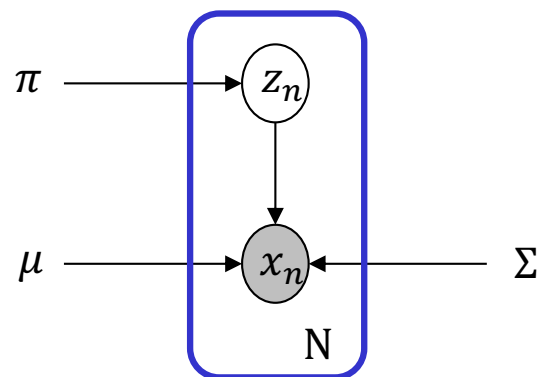
- 3.1 Taxonomy of segmentation algorithms
- 3.2 Validation of segmentation algorithms
- 3.3 Deterministic Filtering & Thresholding Approaches
- 3.4 Probabilistic Imaging Model
- 3.5 Expectation Maximisation for GMM
- 3.6 Image classification with bias field
- 3.7 **Variational Bayes EM**
- 3.8 STAPLE Algorithm

Addressing limitations of EM

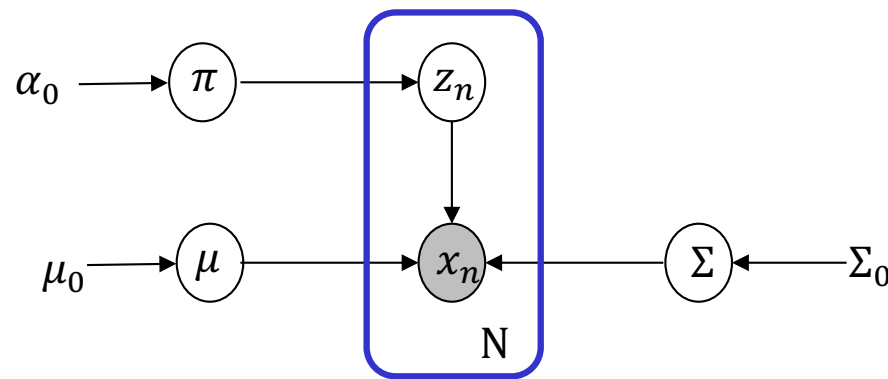
- Limitations on the EM algorithm :
 - 1) Trivial solutions of the maximum Likelihood (Dirac on data points)
 - 2) Non invertibility of Covariance matrices Σ_k
 - 3) Must fix the number of classes prior to the algorithm
- To address those limitations :
add priors on parameters -> Variational Bayes EM

Variational Bayes Gaussian Mixture

- Parameters θ are now random variables
- Define (hyper)prior probabilities $p(\theta)$ to “regularize” their estimated values



Regular Gaussian
Mixture Model



Variational Gaussian
Mixture Model

$\alpha_0, \mu_0, \Sigma_0$ are hyperparameters

How to choose the parameter prior distribution ?

- Convenient choice : “conjugate prior”
 - Lead to closed form expression of posterior

$$p(\theta|X) = \frac{p(X|\theta)p(\theta)}{\int p(X|\theta')p(\theta')d\theta'}$$

- Example :

Likelihood distribution $p(X \theta)$	Parameters	Conjugated Prior $p(\theta)$	Posterior $p(\theta X)$
Gaussian	μ, Σ	Normal x Wishart	Normal x Wishart
Multinomial	π	Dirichlet	Dirichlet

- Existence of conjugated priors for the “exponential family”

Variational Bayes Gaussian Mixture

- Extended EM algorithm :
 - replace maximization of likelihood $p(X|\theta)$ with joint probability $p(X, \theta) = p(X|\theta)p(\theta)$
 - Keep lower bound :
$$\log p(X, \theta) \geq \mathbb{E}_u(\log p(X, Z|\theta)) + H(u) + \log p(\theta)$$
 - Same E-step, but modified M-steps

VBGM : prior on covariance

- Add prior on K inverse Covariance (precision) Matrices Σ_k^{-1} :
 - Objectives :
 - Remove trivial solutions
 - Make Σ_k invertible
 - Wishart Distribution $p(\Sigma_k^{-1}) = \mathcal{W}(\Sigma_k^{-1}; \Sigma_k^0, \beta_k)$ centered on fixed precision (inverse covariance) matrix Σ_0 .
 - New M-Step : $\Sigma_k = \frac{\sum_{n=1}^N u_{nk} (x_n - \mu_k)(x_n - \mu_k)^T + \Sigma_k^0}{\sum_{n=1}^N u_{nk} + 1 + 2\beta_k - (d + 1)}$

VBGM : prior on mixture coefficients

- Add prior on K mixture coefficient $\{\pi_k\} = \pi$:

- Objectives :

- Remove small clusters
- Dirichlet distribution $p(\pi_k) = \mathcal{D}(\pi; \alpha_k)$ where α_k is a positive scalar
- If $\alpha_k \ll 1$ then this prior is sparsity inducing : $p(\pi_k)$ is either 0 or close to 1

- New M-step : $\pi_k = \frac{\sum_{n=1}^N u_{nk} + \alpha_k - 1}{\sum_{k=1}^K \alpha_k + N - K}$

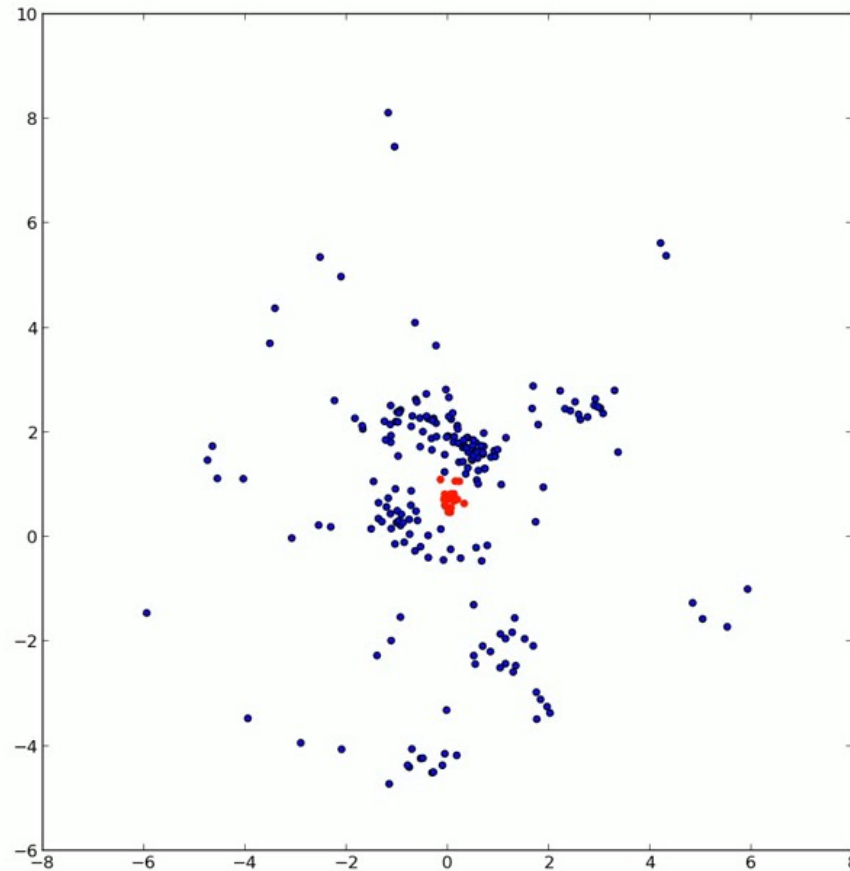
VBGM : prior on mean intensity

- Add prior on K mean values μ_k :
 - Objectives :
 - Constraint mean values to be within certain range given by (μ_k^0, Σ_k^0)
 - Prior is Gaussian distribution characterized by a mean and covariance matrix $p(\mu_k) = \mathcal{N}(\mu_k; \mu_k^0, (\Sigma_k^0)^{-1} \Sigma_k)$
 - New M-step :
$$\mu_k = \frac{\sum_n^N u_{nk} x_n + \Sigma_k^0 \mu_k^0}{\sum_{n=1}^N u_{nk} + \text{tr}(\Sigma_k^0)}$$

VBGM : estimating number of classes

- Initialize GMM with many classes
- Perform VB iterations
 - Dirichlet prior will put move some mixture coefficient to 0 ->
- Remove classes with $\pi_k \approx 0$
- Start again VB iterations until $\pi_k > T$ such that each class has enough samples

Variational Bayes Gaussian Mixture : Example

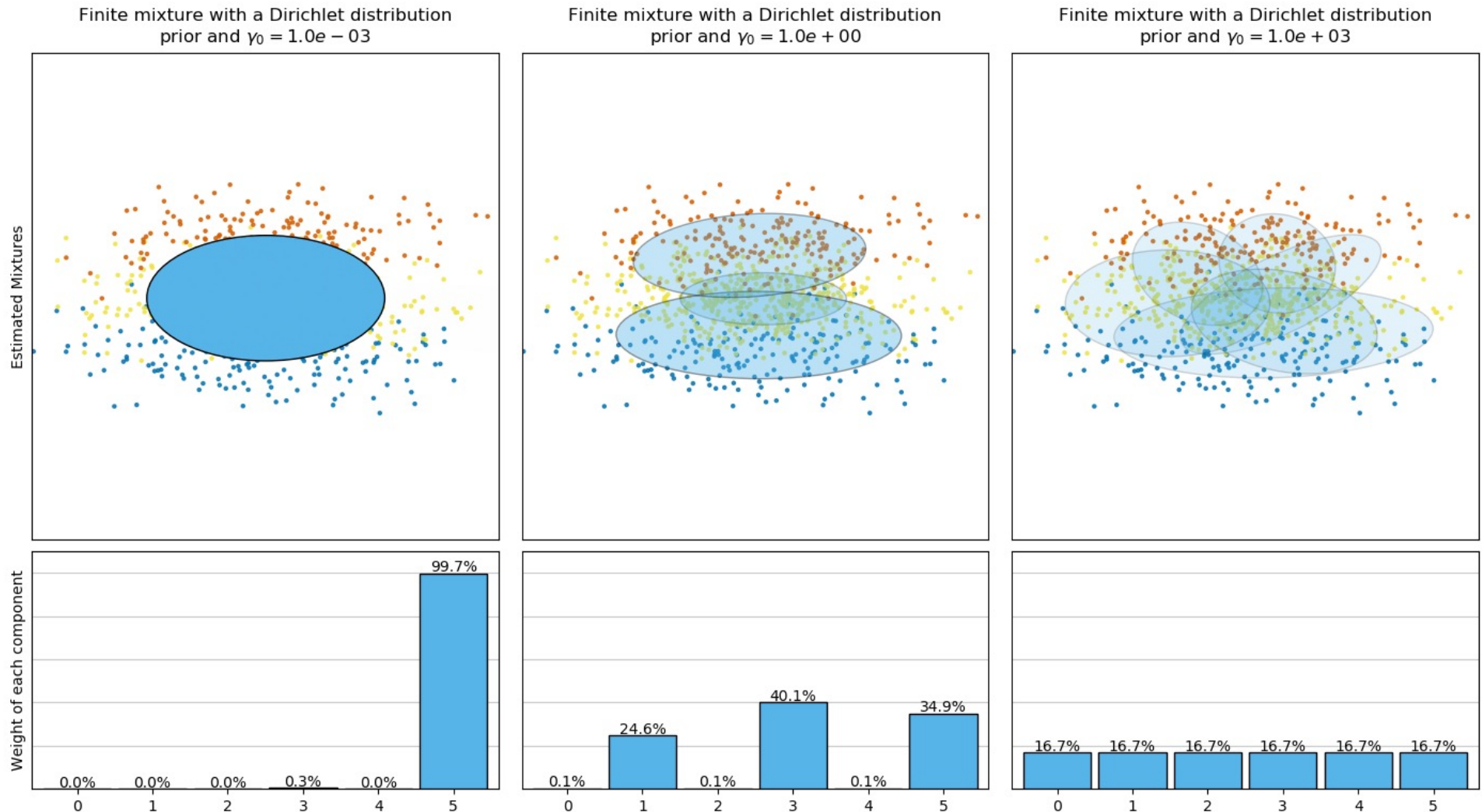


Source : J. Mc Inerney <https://www.youtube.com/watch?v=jijtOcVl0Kw>

Effect of Dirichlet Prior on the number of components

Source : scikit-learn documentation <http://scikit-learn.org/stable/modules/mixture.html#bgmm>

Initial number of clusters is 5



Other extensions of Gaussian Mixture models

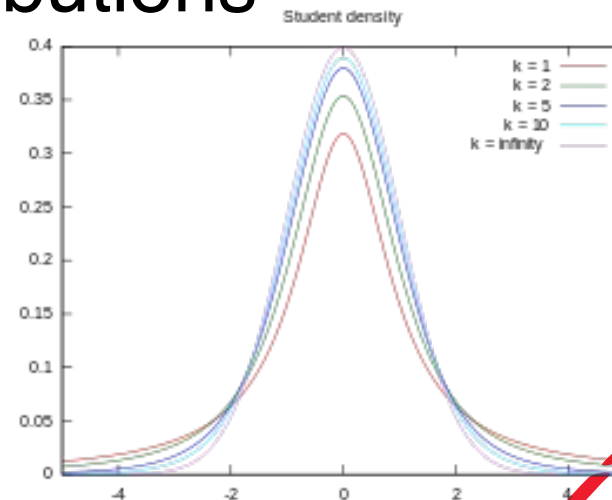
- Dealing with number of classes
 - Use Dirichlet process equivalent to using infinite number of classes
- Robustness to outliers
 - Replace mixture of Gaussians with mixture of Student distributions

$$f(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

ν is the degrees of freedom

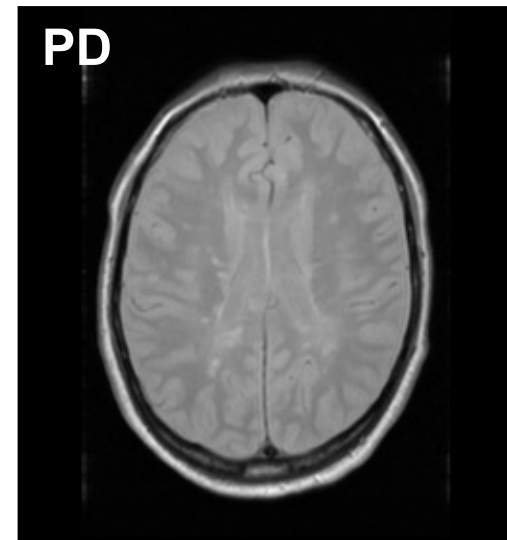
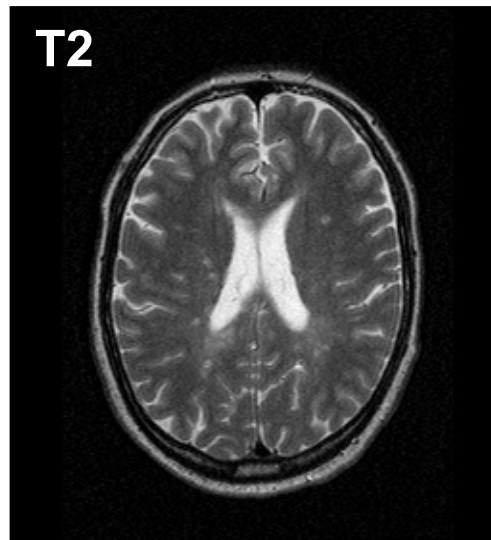
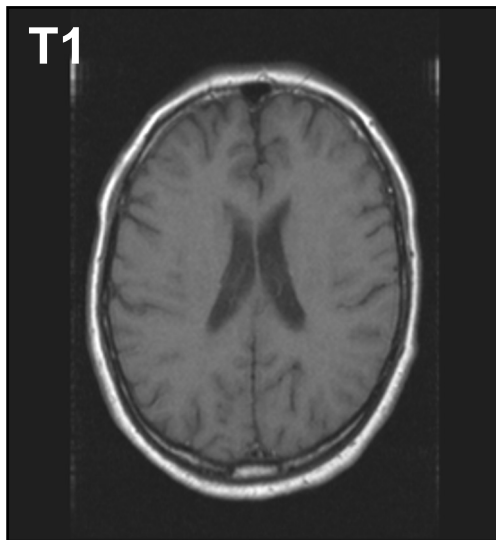
Student tends towards normal distribution

As $\nu \rightarrow \infty$



Other Extensions of EM in Medical Imaging

- Multivariate normal distribution with separate bias fields



References

Paper pioneering EM classification for brain images	WW.M. Wells, W.E.L. Grimson, R. Kikinis, F.A. Jolexz, "Adaptive Segmentation of MRI Data," IEEE Transactions on Medical Imaging , vol 15, no. 4, pp. 429-443, August 1996
Paper including bias field estimation	Van Leemput, K.; Maes, F.; Vandermeulen, D.; Suetens, P., "Automated model-based bias field correction of MR images of the brain," Medical Imaging, IEEE Transactions on , vol.18, no.10, pp.885-896, Oct. 1999.
Other paper with bias field estimation	Guillemaud, R.; Brady, M., "Estimating the bias field of MR images," Medical Imaging, IEEE Transactions on , vol.16, no.3, pp.238-251, June 1997

Medical Imaging : Connexity and Shape Constrained Image segmentation

Hervé Delingette

Epione Team

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4. Connexity and Shape Constrained Image segmentation

- **4.1 Label Connexity Hypothesis : Markov Random Field**
 - Definition of prior
 - Graph cut algorithm
 - Neighborhood EM
 - Grab Cut
- 4.2 Introduction to shape and deformable Models
- 4.3 Snakes algorithm
- 4.4 Level Set Algorithm
- 4.5 Point Distribution Model
- 4.6 Multi-atlas Algorithm

Image Segmentation Approaches

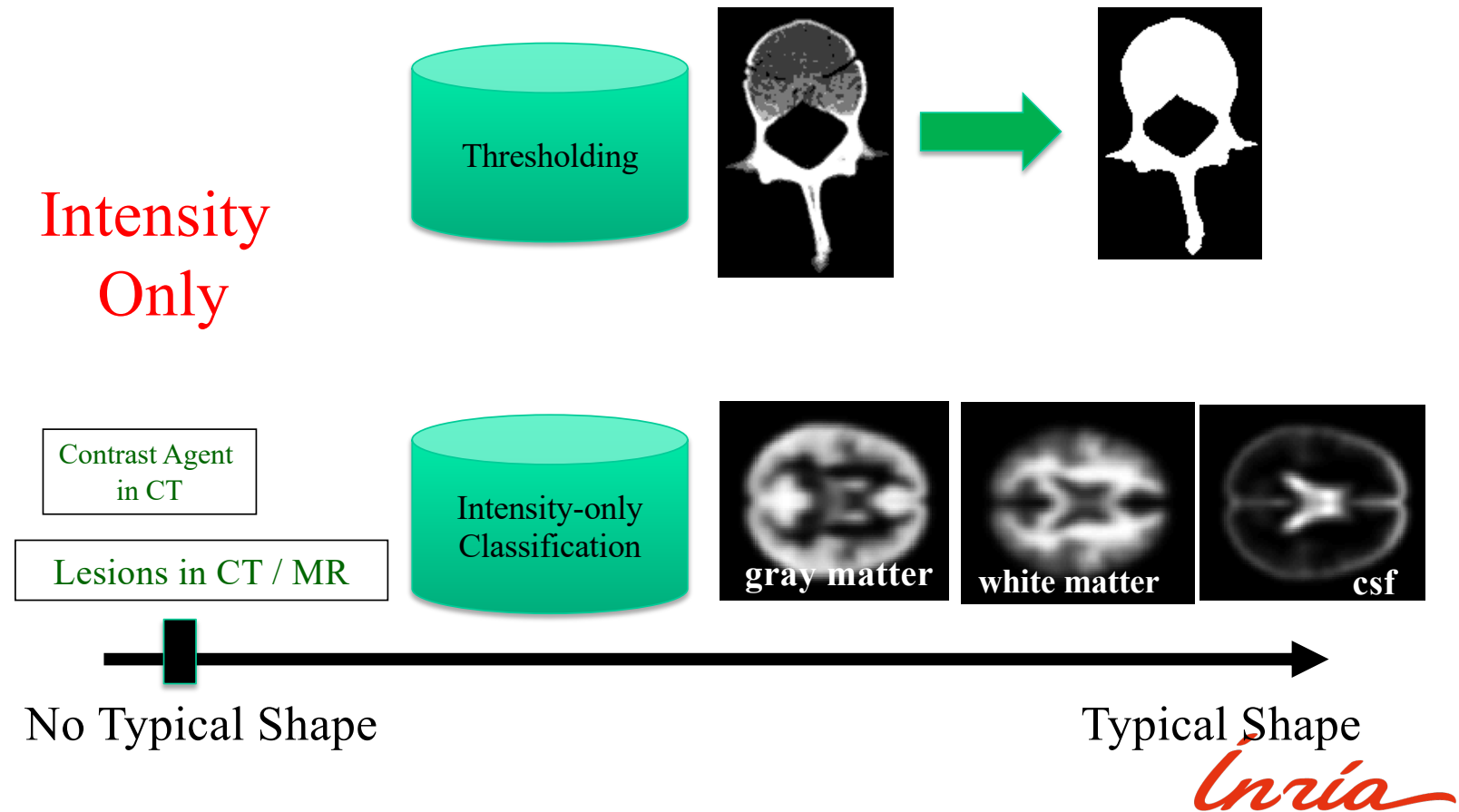
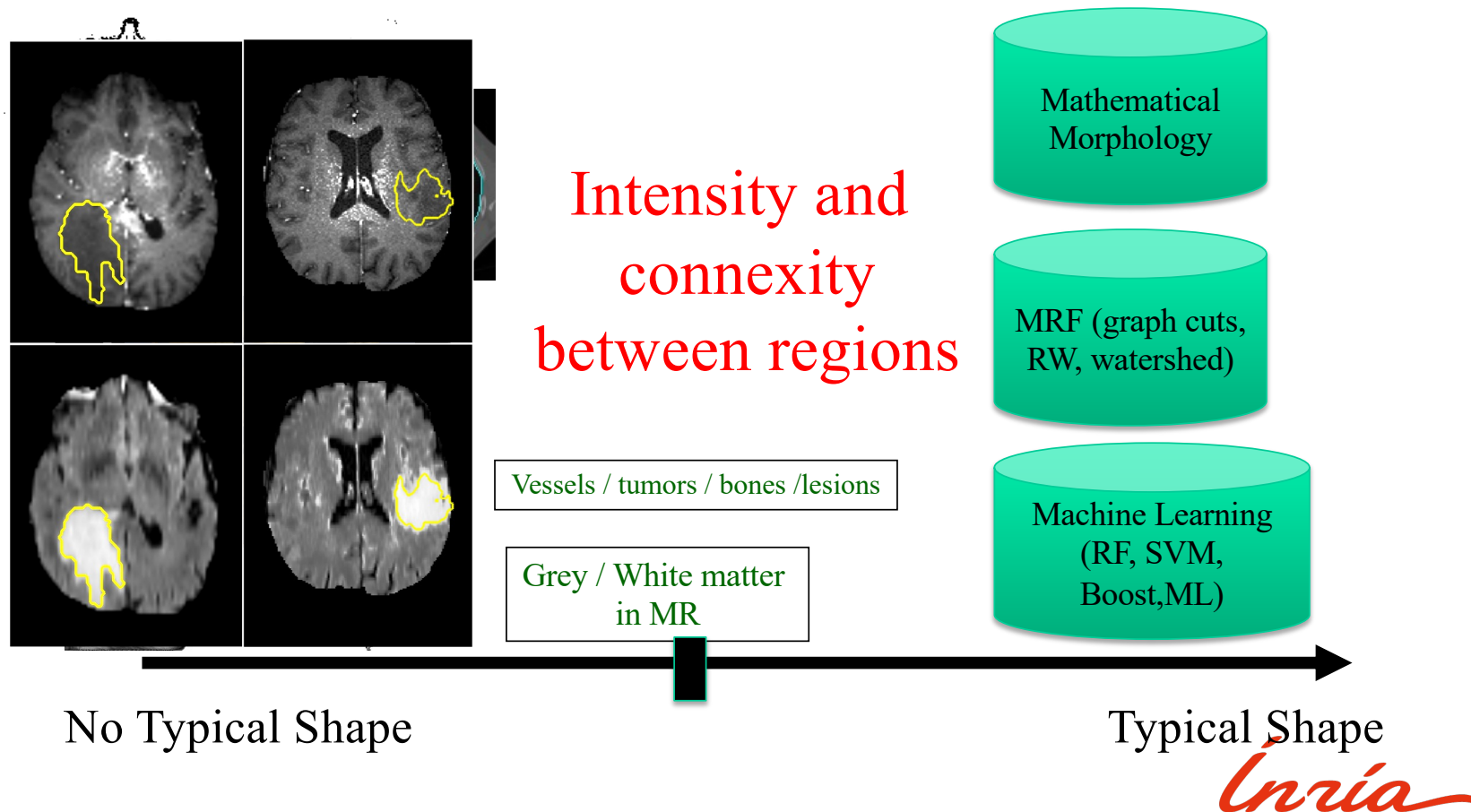
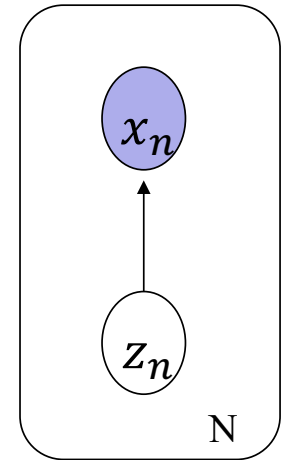


Image Segmentation Approaches



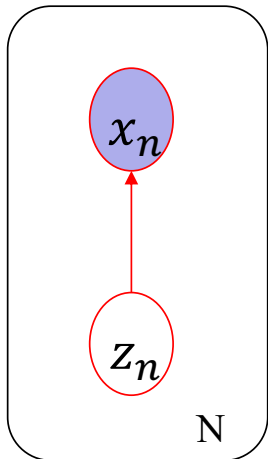
MoG Segmentation Hypothesis

- So far considered independent voxels
 - Z_n variable specifying the class of voxel n
 - X_n variable representing the intensity
- Class membership only dependent on voxel intensity (thresholding)
- But may not be realistic in the presence of noise & partial volume effect

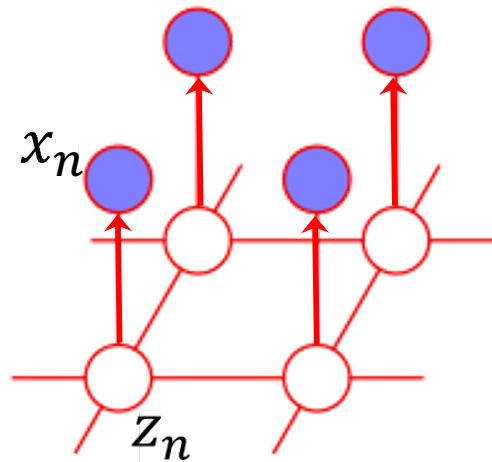


MRF Segmentation Hypothesis

- In Markov Random Fields :
 - Label variables z_n are no longer independent but depend on their neighbors
 - Intensity variables x_n only depends on the class label (variable z_n)



Mixture of Gaussian



Markov Random Field

Markov Random Field

- Intensity prior depends on neighboring values :

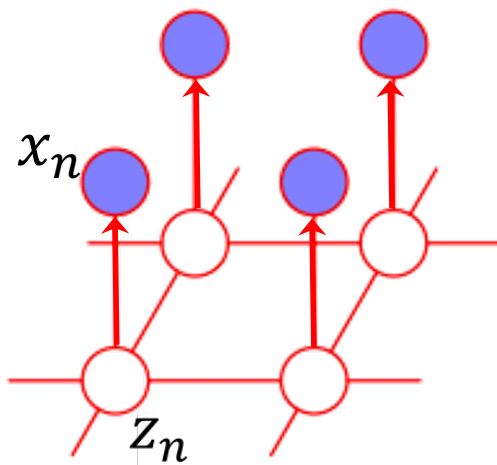
$$p(Z_n | Z_{-n}) = p(Z_n | Z_{N(n)})$$

Label at voxel n

Set of Labels of
all image voxels except
Voxel n

Labels of
Neighboring voxels
Of voxel n

- Graphical Model



x_n are independent only if z_n are known
(conditional independence)

$$p(X) \neq \prod_n p(x_n)$$

$$p(X|Z) = \prod_n p(x_n | z_n)$$

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Challenges in MRF

- Posterior probability is no longer tractable

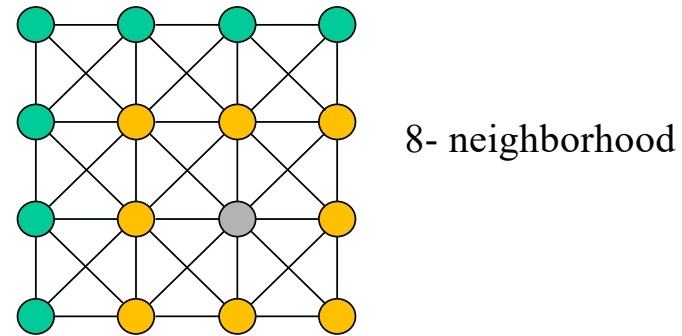
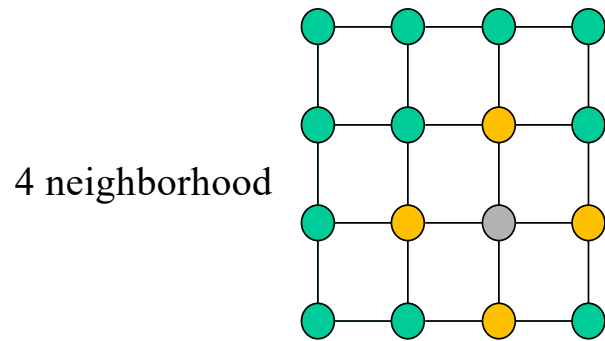
$$p(Z|X) = \frac{p(X|Z)p(Z)}{\sum_{Z'} p(X|Z')p(Z')}$$

$$p(z_n|X) = \underbrace{\sum_{z_1} \sum_{z_2} \dots \sum_{z_{n-1}}}_{\text{Intractable marginalization over } N-1 \text{ term}} \sum_{z_{n+1}} \sum_{z_N} p(Z|X)$$

Intractable sum over 2^N terms

Definition of Label Prior in MRF

- Images seen as Graph



- Label Prior $p(Z)$ depends on neighborhood :
 - 2D images : 4 or 8 neighborhood
 - 3D images : 6, 18 or 26 neighborhood

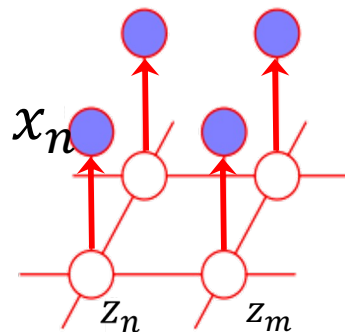
Definition of Label Prior in MRF

- Label prior $p(Z)$ is defined on a graph
4 neighborhood : $p(Z_n|Z_{-n}) = f(Z_{n-1}, Z_{n+1}, Z_{n-R}, Z_{n+R})$

- Hammersley-Clifford theorem gives the expression of $p(Z)$:

- There exists functions ψ and ϕ such that

$$\log p(Z|\theta) = \underbrace{\frac{-1}{T} \sum_{edges(n,m)} \psi(z_n, z_m, \theta)}_{\text{Binary term}} - \underbrace{\frac{1}{T^*} \sum_n \phi(z_n, \theta)}_{\text{Unary term}}$$



$\psi(z_n, z_m, \theta)$ is any function of 2 Binary vectors : it enforces how likely are two labels are different

Unary term

$$\phi(z_n, \theta) = \phi_n$$

Gives how likely voxel n belongs to class k

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Potts Model for Label Prior

- Idea : neighboring voxels should have similar labels.
- Definition Ising when $K=2$:
 - One hot encoding : $Z_n = (Z_{n1}, Z_{n2} \dots Z_{nK})^T$
 - $\psi(z_n, z_m, \theta) = -\sum_{k=1}^K f_{nm} z_{nk} z_{mk}$,
 - In another words :
 - $\psi(z_n, z_m, \theta) = -f_{nm}$ if $Z_n = Z_m$ and $\psi(z_n, z_m, \theta) = 0$ if $Z_n \neq Z_m$,
- Alternative 1 : $\psi(z_n, z_m, \theta) = f_{nm} \|Z_n - Z_m\|^2$
- Coefficient definition : neighboring voxels having similar intensity should have the same labels.

$$f_{nm} = \exp -\beta (x_n - x_m)^2$$



Joint Probability in MRFs

- Definition of joint probability :

- $p(X, Z|\theta) = p(Z)p(X|Z)$

- Log joint probability

$$\Lambda(Z, \theta) = \log p(X, Z|\theta) = \log p(Z|\theta) + \log p(X|Z, \theta)$$

Conditional independence



$$\Lambda(Z, \theta) = \log p(Z|\theta) + \sum_n \log p(x_n|z_n, \theta)$$

Categorical variable



$$\Lambda(Z, \theta) = \log p(Z|\theta) + \sum_n \sum_k z_{nk} \log p(x_n|z_{nk} = 1, \theta)$$

Energy

$$-\Lambda(Z, \theta) = \underbrace{\frac{1}{T} \sum_{edges(n,m)} \psi(z_n, z_m, \theta)}_{\text{Binary term}} + \underbrace{\frac{1}{T^*} \sum_n \phi(z_n, \theta) - \sum_n \sum_k z_{nk} \log p(x_n|z_{nk} = 1, \theta)}_{\text{Unary terms}}$$

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Algorithms for solving MRF

- Many existing algorithms :
 - 1) **Graph cut Algorithm** :
 - Fast
 - solve for hard memberships z_{nk}
 - Unique solution for $K=2$ if some constraints on f_{nm} are met
 - Several extensions for $K>2$
 - 2) **Neighborhood EM**
 - solve for soft memberships $p(z_n|x_n)$
 - Simple Extension of GMM
 - Fixed point Iterative method
 - 3) **Grab Cut**

Graph cuts

- Binary case & Ising model :

- 2 labels case $y_i \in \{0,1\}$

- Minimize energy :

$$E(Y) = \sum_{i,j} c_{ij} y_i (1 - y_j) + \sum_i d_i y_i, \text{ with } d_i > 0$$

- Submodular constraint for unique solution

$$c_{ij} + c_{ji} \geq 0$$

- Minimize $E(Y)$

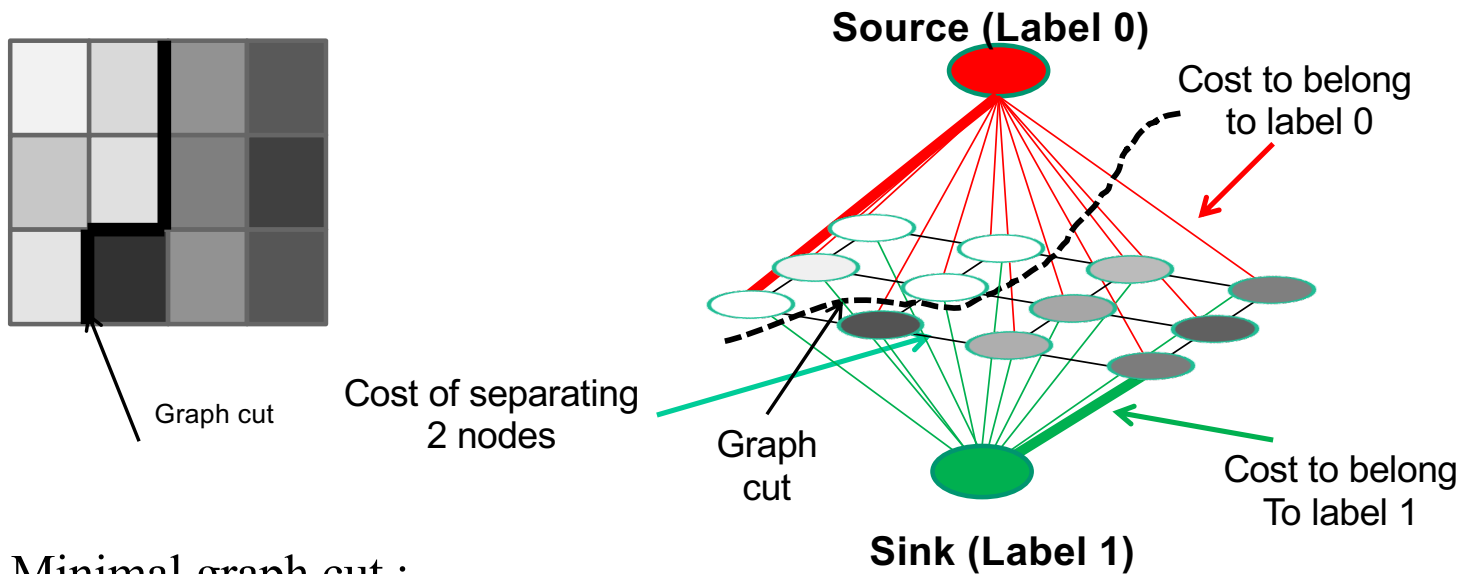


Minimize a graph cut

Combinatorial problem

D.M. Greig, B.T. Porteous and A.H. Seheult (1989), *Exact maximum a posteriori estimation for binary images*,
Journal of the Royal Statistical Society Series B, **51**, 271–279.

Graph Cut



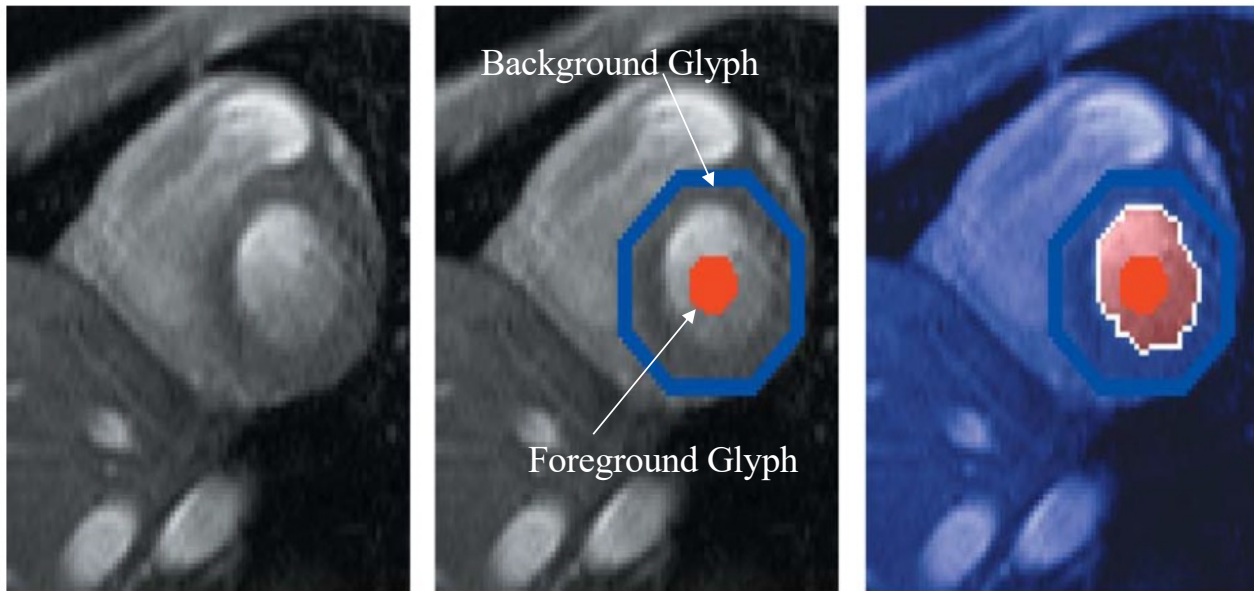
- Minimal graph cut :
 - Set of edges whose removal create several connected components:
 - Cost of a cut :

$$cut(A, B) = \sum_{p \in A, q \in B} c_{p,q}$$

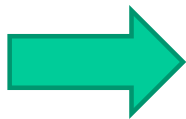


Maximize the flux between the source and the sink nodes

Interactive Segmentation Algorithm



Source : Boykov & Gareth Funka-Lea Graph Cuts and Efficient N-D Image Segmentation



Manual glyph from user to guide
segmentation

Graph cut Segmentation

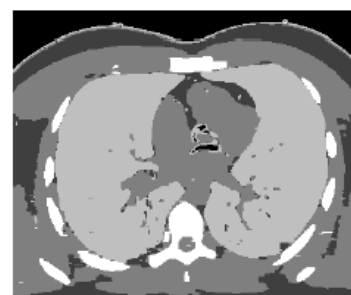
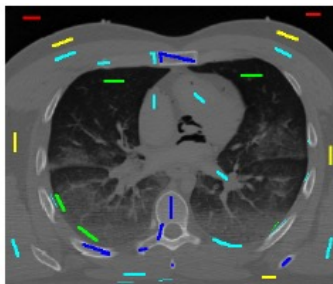
- Combinatorial algorithm for graph cut :

Ford & Fulkerson Algorithm (1951)

BoyKov & Kolmogorov Algorithm (2004)

Y. Boykov and V. Kolmogorov. An experimental comparison of min-cut/max-flow algorithms for energy minimization in vision. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 26(9):1124–1137, September 2004.

- Multi Label Segmentation with α -expansion algorithm [Veksler 99] [Boykov 99]



R. Kéchichian, S. Valette, M. Desvignes, R. Prost: **Efficient multi-object segmentation of 3D medical images using clustering and graph cuts**. ICIP 2011

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Neighborhood EM

- Hypothesis :

- Posterior probability $p(z_n|X)$ is intractable therefore estimate an approximation
- Each tissue class is represented by a Gaussian distribution

$$p(x_n|z_{nk} = 1) = \mathcal{N}(x_n|\theta_k)$$

- The label prior is a Potts model and global prior per class

$$\log p(Z) = -\frac{\beta}{2} \sum_k \sum_{edges(m,n)} c_{nm} z_{nk} z_{mk} + \sum_n \sum_k \pi_k z_{nk}$$

C. Ambroise , M. Dang , G. Govaert: Clustering of Spatial Data by the EM Algorithm. In geoENV
I-Geostatistics for Environmental Applications (1997), pp. 493-504.



Mean Field approximation

- A.ka Variational Bayes approach

- Look for an approximation of posterior parameters as product $q(Z) = \{q_n\}$ of factorized terms $p(Z = \{z_n\}|X) \approx \prod_n q_n(z_n)$

- Therefore NK unknown q_{nk} s.t

$$q_n(z_n) = \sum_k q_{nk} z_{nk} \quad \& \quad \sum_k q_{nk} = 1 = \sum_{z_n} q_n(z_n)$$

- Find the set q which minimizes the Kullback Leibler divergence between q and true posterior $p(Z|X)$

Mean Field Criterion

- Reminder EM criterion for GMM :

- Maximize $F(\pi, \theta, u)$

$$F(\pi, \theta, u) = L(\pi, \theta) - D_{KL}(u || p(z|x)) = Q(\theta, u) + H(u)$$

- Evidence Lower bound :

$$D_{KL}(q || p(Z|X)) = -\log p(X) - \mathbb{E}_q (\log p(X, Z)) - H(q)$$

- Neighborhood EM criterion same as GMM but with additional term $R(q)$

$$\text{minimize } D_{KL}(q || p(Z|X)) = -H(q) + R(q) - Q(q) + \log p(X)$$

- Where $R(q) = \frac{\beta}{2} \sum_k \sum_{edges(n,m)} c_{nm} q_{nk} q_{mk}$

Neighborhood EM

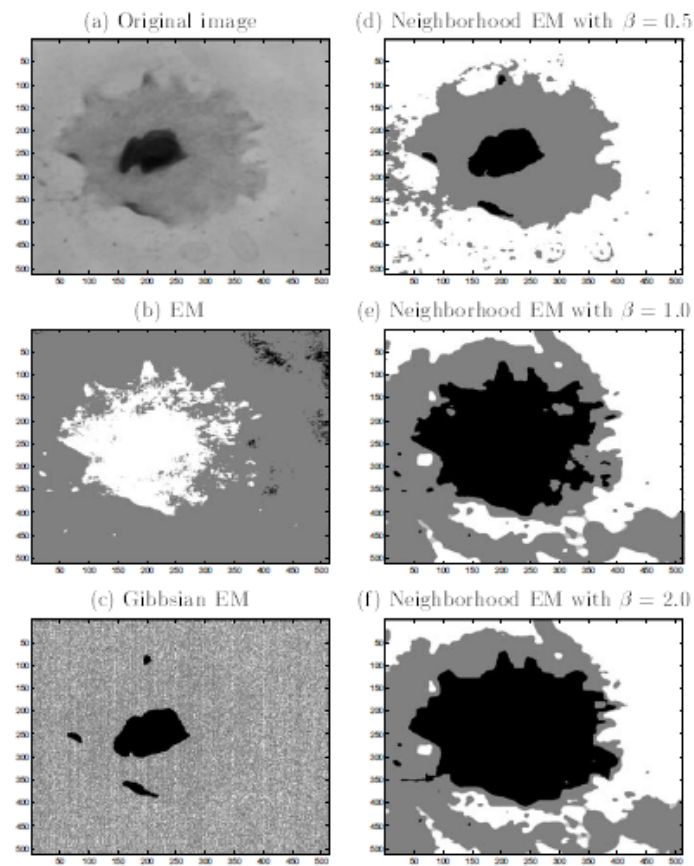
- Only E-step changed compared to regular EM for GMM
- New E-step :
 - Fixed point iteration

$$q_{nk} = \frac{\pi_k \mathcal{N}(x_n | \theta_k) \exp \beta \sum_m c_{mn} q_{nm}}{\sum_l \pi_l \mathcal{N}(x_n | \theta_l) \exp \beta \sum_m c_{mn} q_{nm}}$$

- Same M-step

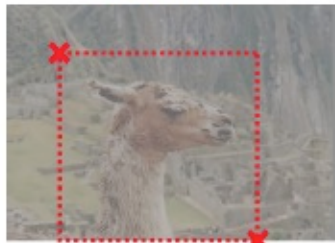
$$\mu_k = \frac{\sum_{n=1}^N q_{nk} x_n}{\sum_{n=1}^N q_{nk}} \quad \Sigma_k = \frac{\sum_{n=1}^N q_{nk} (x_n - \mu_k)(x_n - \mu_k)^T}{\sum_{i=1}^N q_{nk}} \quad \pi_k = \frac{1}{N} \sum_{n=1}^N q_{nk}$$

Neighborhood EM



Grab Cut

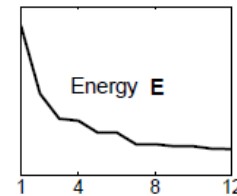
- Algorithm combines :
 - Model intensity of foreground and background as mixture of Gaussians (vs one Gaussian for each class)
 - Iterate between :
 - hard segmentation using graph cuts
 - Estimation of Gaussian components



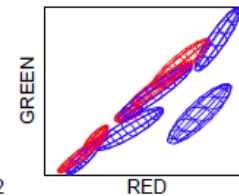
Input + bounding
box



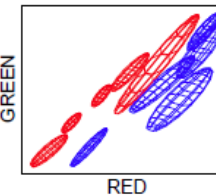
Segmentation
(f)



(a)



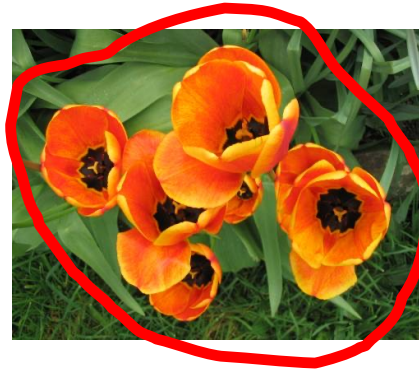
(b)
Initial GMM



(c)
Final GMM

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Grab Cut Examples



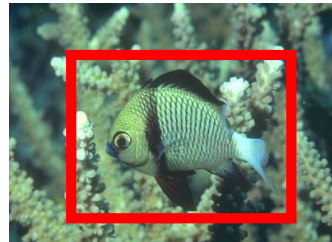
Available in MS Office !!

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Difficult Examples

Camouflage &
Low Contrast

Initial
Rectangle



Initial
Result



Fine structure



Harder Case



Grabcut: Interactive foreground extraction using iterated graph cuts, Carsten Rother, V. Kolmogorov, Andrew Blake, Siggraph 2004

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